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A novel center-based deep contrastive metric learning method for the detection of polymicrogyria in pediatric brain MRI

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ABSTRACT

Polymicrogyria (PMG) is a disorder of cortical organization mainly seen in children, which can be associated with seizures, developmental delay and motor weakness. PMG is typically diagnosed on magnetic resonance imaging (MRI) but some cases can be challenging to detect even for experienced radiologists. In this study, we create an open pediatric MRI dataset (PPMR) containing both PMG and control cases from the Children's Hospital of Eastern Ontario (CHEO), Ottawa, Canada. The differences between PMG and control MRIs are subtle and the true distribution of the features of the disease is unknown. This makes automatic detection of potential PMG cases in MRI difficult. To enable the automatic detection of potential PMG cases, we propose an anomaly detection method based on a novel center-based deep contrastive metric learning loss function (cDCM). Despite working with a small and imbalanced dataset our method achieves 88.07% recall at 71.86% precision. This will facilitate a computer-aided tool for radiologists to select potential PMG MRIs. To the best of our knowledge, our research is the first to apply machine learning techniques to identify PMG solely from MRI.

Our code is available at: https://github.com/RichardChangCA/Deep-Contrastive-Metric-Learning-Method-to-Detect-Polymicrogyria-in-Pediatric-Brain-MRI.

Our pediatric MRI dataset is available at: https://www.kaggle.com/datasets/lingfengzhang/pediatric-polymicrogyria-mri-dataset.

1. Introduction

Polymicrogyria (PMG) is a cortical malformation characterized by irregular thickened gray matter in the cerebral cortex, with numerous small gyri, shallow sulci and loss of gray white matter differentiation (see Fig. 1). Polymicrogyria is one of the most common malformations of cortical development and is most often related to genetic disorders although it can be caused by congenital viral infection (Barkovich, 2010). PMG is a heterogeneous condition which can have variable distribution, extent, and severity (Barkovich, 2010). Experienced neuroradiologists can accurately diagnose PMG using magnetic resonance imaging (MRI) but focal or more subtle PMG can be missed. Less experienced readers may not detect PMG with the same accuracy as it can be subtle, and computer aided diagnosis would be useful here.

Machine learning has been applied to cortical malformations in the pediatric brain, mainly in the context of fetal imaging (Attallah et al., 2020) and epilepsy (Sone and Beheshti, 2021) and may contribute to computer-aided diagnosis. There is however a lack of literature applying machine learning and in particular deep learning to the detection of polymicrogyria.

The disease is rare in most clinical settings, while collecting normal pediatric brain MRI is relatively easy. We collected and annotated a small pediatric PMG MRI dataset (23 patients) from the Children's Hospital of Eastern Ontario (CHEO), Ottawa, Canada. This pediatric polymicrogyria MRI (PPMR) dataset is imbalanced because the majority of MRI are control and even for PMG samples the majority of MRI slices do not show evidence of PMG. We believe the composition of the PPMR dataset is a realistic reflection of the data likely to be encountered in clinical practice. In this paper, we introduce the PPMR dataset in detail. Because of the nature of PMG and the difficulty in collecting data, this dataset is unfortunately small and lacks the diversity of PMG. As it is difficult to evaluate testing results on such a dataset, we also introduce a modified CIFAR-10 dataset. This modified CIFAR-10 dataset mimics the challenges of our PPMR dataset, including the fact that the minority

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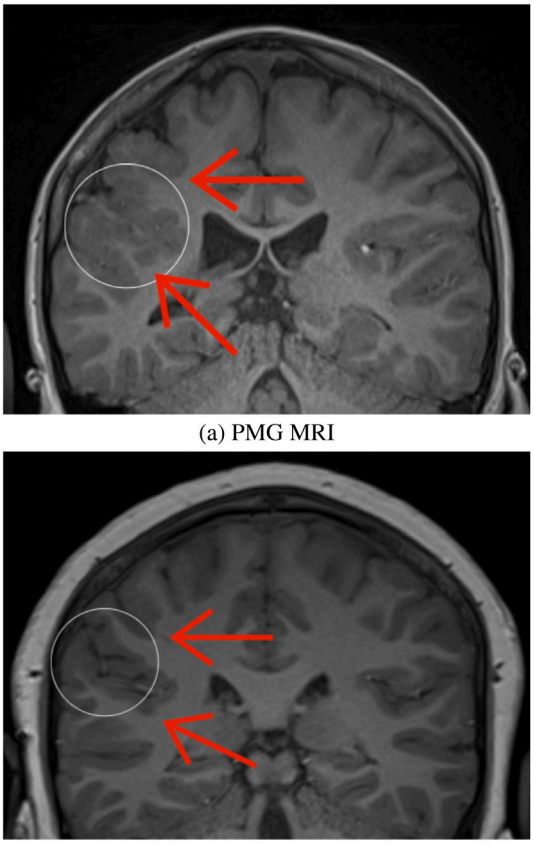


(a) PMG MRI

(b) Control MRI

Fig. 1. Difference between PMG and normal brain on coronal T1 weighted MR images. The white circle on the first image shows thickened, irregular gray matter with numerous small gyri and shallow sulci, compared to normal gray matter in the white circle on the second image. The interface between cortical gray matter and white matter is irregular in PMG (red arrows on the first image) and smooth in normal brain (red arrows on the second image).

class samples in the training data are not representative and that the whole dataset is imbalanced.

Deep learning models trained with cross-entropy-based loss functions on small imbalanced datasets, perform poorly when applied to real-word out-of-distribution samples. Anomaly detection has the potential to mitigate this issue. This motivates the design of our novel cDCM loss function that combines anomaly detection and binary classification. Each input MRI is mapped to a latent representation where our loss function can "cluster" normal samples together, because normal images share similar features, and push anomaly samples further away from the "cluster". During inference, according to the idea of supervised contrastive learning (Khosla et al., 2020), samples that have similar features as normal samples in the training data can be classified as normal; on the other hand, samples that have dissimilar features from normal samples in the training data can be classified as anomalies.

Our novel cDCM loss function can partly mitigate the problem of generalization in deep learning. In this paper, we compare our novel cDCM loss function to the Deep SAD loss function (Ruff et al., 2019), triplet loss (Ju et al., 2020) and cross-entropy-based loss functions to demonstrate the benefits of our novel cDCM loss function. Our proposed cDCM loss function overcomes the challenge of determining a decision threshold that cannot be reliably obtained from training and validation data when using cross-entropy-based loss functions. We employed ResNet50 (He et al., 2016) as the convolutional neural network backbone for feature extraction from MRI images.

In summary, the main contributions of our research are listed below:

- We propose a novel center-based deep contrastive metric learning loss function (cDCM) to work with a small and imbalanced dataset for which some testing data features do not appear in the training data.
- Our method integrates the strengths of supervised binary classification and anomaly detection and to the best of our knowledge, we are the first to apply deep contrastive learning to PMG MRI classification.
- We make our pediatric PMG MRI dataset (a.k.a. PPMR dataset) from the Children's Hospital of Eastern Ontario (CHEO), Ottawa, Canada available for research.

The remainder of this paper is organized as follows: In Section 2, we give a literature review, including machine learning in PMG images, classification of brain MRI, anomaly detection in medical imaging, and deep metric learning and deep contrastive learning. In Section 3, we give a detailed illustration of our proposed novel cDCM loss function and model structure. In Section 4, we introduce two datasets: a modified CIFAR-10 dataset and our PPMR dataset, and also describe training details. In Section 5, we show the results of substantial experiments on these two datasets and also analyze and discuss the results of these experiments. Finally, we give a conclusion of our research and also consider some future works.

2. Related work

2.1. Machine learning in polymicrogyria images

There is limited literature about applying machine learning methods on detecting pediatric PMG from MRIs (Attallah et al., 2018), however, there is work on the pediatric brain which can also inform research on pediatric PMG. Pediatric brain segmentation (Devi et al., 2015; Mostapha and Styner, 2019) and brain reconstruction (Kim et al., 2023) are of interest. Deep learning has been applied successfully to these tasks, e.g.: Mostapha and Styner (2019) addressed brain tissue segmentation tasks of infants and Kim et al. (2023) improved image quality in pediatric brain MRI reconstruction. It has also been noted that processing adult and pediatric brain MRI require different techniques. For example, Devi et al. (2015) conducted a literature review on neonatal brain MRI segmentation, pointing out that some adult brain processing techniques are not applicable to neonatal brains due to the inverted contrast between gray and white matter.

Attallah et al. (2018) proposed a machine learning pipeline for fetal brain MRI classification. This method can predict several brain abnormalities, including PMG, before the infant with fatal brain abnormalities is born. This machine learning pipeline consists of four steps: region of interest (ROI) segmentation, contrast enhancement, feature extraction, and classification by linear discriminate analysis (LDA), support vector machine (SVM), K-nearest neighbor (KNN) and ensemble classifiers. Attallah et al. (2020) proposed an improved method that combines CNN feature extraction based on transfer learning and conventional machine learning classifiers to detect fatal brain MRIs. We do not compare our method to Attallah et al. (2020), despite the fact that their method led to 88.6% accuracy and 94% AUC on their MRI data, because transfer learning did not extract useful features from our PMG images in our tests. Hence, we could not apply the approach of Attallah et al. (2020) as transfer learning is required.

Płoński et al. (2017) collected 740 features, such as cortical thickness and surface area, from T1-weighted MRIs and then used feature selection methods and a logistic regression classifier to detect congenital brain malformations including PMG that lead to dyslexia. The best results are 66% AUC and 65% accuracy after 10-fold cross-validation. They demonstrated the main reason leading to these relatively poor results is that dyslexia is a heterogeneous syndrome and there is little consistency in univariate gray matter.

However, these methods above are not specific to the detection of PMG on MRIs, because they applied their methods to a dataset which only includes a small portion of PMG. In addition, these methods have not shown they have the ability to deal with the problem of imbalanced datasets and generalization to unseen features.

2.2. Classification of brain MRI

While many approaches for medical image classification exist, we focus our brief review on related deep learning methods.

Medical brain data is typically volumetric. Since 3D brain MRIs include more information compared to 2D brain MRIs, some researchers applied deep learning models to 3D brain MRIs directly and achieved good results when they have sufficient 3D volumetric data (Korolev et al., 2017; Wegmayr et al., 2018). Korolev et al. (2017) proposed an end-to-end 3D CNN architecture to classify Alzheimer's disease from normal controls on the 3D Alzheimers Disease National Initiative (ADNI) dataset. Wegmayr et al. collected 3D brain MRIs from various sources to build a large 3D brain dataset and customized a 3D CNN architecture to classify several neurodegenerative diseases (Wegmayr et al., 2018). However, in our PPMR dataset, we only have 23 PMG brains in total, which is not sufficient for training a 3D CNN model. We can regard each PMG brain as a series of PMG slices; then the number of data samples is sufficient to train a deep learning model.

In medical imaging, it is difficult to acquire a sufficient number of images because of a limited number of patients and because of privacy concerns (Greenspan et al., 2016). Deep learning models are easily overfitting to the training data if they are trained on relatively small datasets (Pasupa and Sunhem, 2016). Consequently, deep learning models will lack the ability to generalize on unseen data. CNNs as feature extractors with conventional machine learning models can improve generalization (Deepak and Ameer, 2019). However, in our case, deep feature extractors and traditional image processing methods do not extract essential features from our PPMR data because the differences between normal and PMG MRI are subtle.

2.3. Anomaly detection in medical imaging

Pathologies in medical images are often regarded as rare deviance (anomalies) because the majority of medical images are healthy (or normal) samples (Tschuchnig and Gadermayr, 2022). Unlike binary classification, anomaly detection can deal with a relatively imbalanced dataset where normal samples are the majority and the diversity of anomaly samples is limited.

While semi-supervised and unsupervised deep learning methods are common in anomaly detection (Tschuchnig and Gadermayr, 2022), their utility may be limited for medical images which have complex or even indistinguishable features (Gao et al., 2020), such as PMG images. Therefore, exploring anomaly-supervisory signals may help anomaly detection (Pang et al., 2021). Ruff et al. introduced a deep semi-supervised anomaly detection (Deep SAD) (Ruff et al., 2019) which showed that adding a few labeled anomaly samples during training can achieve better results than the deep support vector data description (Deep SVDD) (Ruff et al., 2018), which is trained only on normal samples. Ding et al. (2022) proposed a novel open-set supervised anomaly detection method to detect seen anomalies and unseen anomalies. They created some pseudo anomaly samples which have similar features as real anomaly samples but are dissimilar to normal samples. However, in our PMG detection, real anomaly samples are close to normal samples because the differences between PMG MRI and normal MRI are subtle. Hence, generating pseudo anomaly samples for our PMG detection may not be appropriate.

Kawachi et al. (2018) defined the normal distribution and the anomaly distribution as analogous to a set and its complementary set, and adapted these two complementary sets into a variational autoencoder for performing supervised anomaly detection. However, autoencoder-based anomaly detection methods generally demand a larger number of parameters, double that of the encoder alone. This could become impractical when employing complex encoder architectures, such as ResNet50. For our polymicrogyria images, a complex encoder is required to extract sufficient feature information due to the high complexity and heterogeneity of the data.

Furthermore, Defard et al. (2021) proposed a new approach for patch distribution modeling, known as 'PaDiM'. They leveraged an ImageNet pretrained CNN model to extract image features, which were then transformed into patch embedding vectors. For each patch location, the probabilistic distribution was computed based on all normal training samples, and anomaly samples were identified as deviations from the density of the distribution. No deep learning training process was required because the method relies heavily on features extracted from the ImageNet pretrained CNN model. For each patch location, anomalous parts were detected enabling anomaly localization and segmentation. For anomaly detection, the highest anomaly score across all patches within an image served as the anomaly score of the whole image. Reiss et al. (2021) also adapted pretrained features for anomaly detection. Their framework consists of 3 key steps: 1. They selected an ImageNet pretrained CNN model to serve as the initial feature extractor. 2. They noticed that these features need to be further adapted for anomaly detection due to feature deterioration. They incorporated a compactness loss function, identical to the one used in Deep SVDD (Ruff et al., 2018). They also used elastic weight consolidation (EWC) (Kirkpatrick et al., 2017), inspired by continual learning, to update model weights. 3. Finally, they employed K-Nearest Neighbors (KNN) for anomaly scoring. Since these two methods have all demonstrated superior performance compared to many state-of-the-art methods, we present a comparison of our method with PaDiM and PANDA in Section 5.4.

2.4. Deep metric learning and deep contrastive learning

Deep metric learning (Kaya and Bilge, 2019) and deep contrastive learning (Khosla et al., 2020) share similar ideas. Both of them learn embedding representations from inputs. However, deep metric learning was inspired by traditional metric learning methods (Kulis et al., 2013), i.e., t-SNE (Van der Maaten and Hinton, 2008), K-nearest neighbors (Peterson, 2009), and relevant variants. Metric learning maps input data samples into embedding representations and then calculate distances between these embeddings for clustering or classification tasks. Deep metric learning replaces sophisticated manually designed mapping functions with deep neural networks. Deep contrastive learning is a subset of deep metric learning where methods learn embedding representations in a contrastive fashion (Chopra et al., 2005). It requires that similar data samples (data samples from the same class) are close to each other in the embedding space and dissimilar data samples (data samples from different classes) are far away from each other in the embedding space. Han et al. (2021) used a contrastive learning method to pre-train a feature extractor in a self-supervised way and added a classification head to fine-tune the whole model for pneumonia detection on chest X-ray images. Similarly, Chen et al. (2021) also used a contrastive loss to train an encoder. But they adopted the pre-trained encoder for classification using a few-shot learning method to diagnose COVID-19 based on chest CT images. More similar works (Shurrab and Duwairi, 2021; Sowrirajan et al., 2021; Sriram et al., 2021; Azizi et al., 2021) used contrastive learning methods to learn better feature representations from medical images.

Moreover, Tian et al. (2021) applied contrastive distribution learning with two pretext learning tasks for unsupervised medical anomaly detection. The algorithm proceeds by contrastive pretraining followed by anomaly detection based on pretrained embedding. The contrastive pretraining model structure includes image-level and patch-level encoders. For each level encoder, they constructed image pairs through data augmentation and encouraged these pairs to be closer to each

other than to other images. Recently, Tian et al. introduced pseudo multi-class strong augmentation via contrastive learning PMSACL (Tian et al., 2023). The main advantage of PMSACL compared to their previous work is incorporating medical imaging domain knowledge into the training algorithm. Different level data augmentations were added to the original image to simulate diseases of varying severity, followed by a tailored multi-center contrastive learning method applied for pretraining. Both of these two methods employed contrastive learning for unsupervised pretraining, i.e. two-stage algorithms and different from our method because our cDCM loss directly use contrastive learning to distinguish normal and anomaly samples, making it a one-stage supervised learning method. Yang et al. (2022) introduced a one-stage prototype-aware contrastive learning method called ProCo. This approach utilized contrastive learning as an auxiliary task to partly address long-tailed medical imaging classification issue while still employing a cross entropy loss function for the classifier.

In addition, Ju et al. (2020) employed an iterative process to extract reliable normal and potential anomaly samples from unlabeled data. They also incorporated a small number of labeled positive samples to construct triplets for deep metric learning-based anomaly detection, called "PUMAD" . Du and Zhang (2014) employed manifold features to identify potential normal and anomaly parts from image pixels, and subsequently also constructed triplets to detect anomalies. The construction of triplets in deep metric learning typically requires two samples from the normal class to be semantically similar, while a pair consisting of one normal and one anomaly sample should be semantically different. However, in the case of our PPMR dataset, it is possible for two samples from the normal class to be heterogeneous. For instance, a special scenario could involve one normal image from the forebrain of one patient and another normal image from the hindbrain of another patient. In this case, the two images are heterogeneous despite both being labeled as normal. It is also worth noting that it is also possible for two samples from different classes to be semantically similar, such as two consecutive MRI images from one patient's brain. We compare the cDCM loss with the triplet loss in the modified CIFAR-10 dataset for a comprehensive study but we note that employing triplet loss-based deep metric learning methods may not be suitable for our PPMR dataset. In conclusion, to address the issues mentioned above, we propose a center-based deep contrastive metric learning approach.

3. Proposed method

Cross-entropy-based loss functions have several limitations: they easily lead to overfitting during training on small datasets; they tend to predict testing samples as the majority class; and they lack generalization when encountering unseen features. To mitigate these issues, we propose a novel cDCM loss function for center-based deep contrastive metric learning and utilize a ResNet50 backbone for PMG detection in pediatric brain MRIs. Our novel cDCM loss function can improve model generalization, and the decision boundary of this loss function tends to produce high recall with acceptable precision over two datasets: the modified CIFAR-10 and the PPRM. Since the goal of our research is to design a computer-aided diagnosis method to identify possible PMG, and radiologists should focus on images predicted as possible PMG, recall is more important than precision in our setting. We introduce our novel cDCM loss function next.

3.1. Loss function

Inspired by the loss function of Deep SAD (Ruff et al., 2019) and the hinge loss (Wu and Liu, 2007), we propose a novel center-based deep contrastive metric learning loss function (cDCM), which has a clear predefined decision boundary to help make the final decision. Our deep learning model f maps each input image to an n-dimensional latent representation.

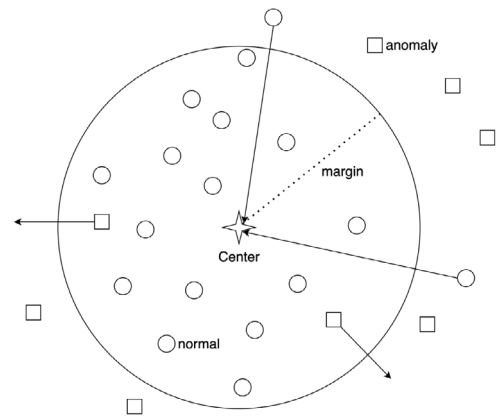


Fig. 2. Illustration of our novel cDCM loss function, the four-pointed star represents the center c, anomaly samples in the latent representation are marked with a square, normal samples in the latent representation are represented by a small circle, and the large circle represents the decision boundary.

The basic idea of our novel cDCM loss function is shown in Fig. 2. This loss can force negative (normal) samples to be close to the center c, and it can also push positive (anomaly) samples outside the circular margin centered at c.

We use the Euclidean norm to measure the distance between the latent representations of samples and the center c, i.e., $d_i = \|l_i - c\|$, where l_i is the latent representation of sample i, and d_i is the Euclidean norm between the latent representation and the center c. To make sure the majority of positive (anomaly) samples are not close to the decision boundary (margin), we add the term $\frac{1}{1+e^{d_i-m}}$, which increases the gradients for positive (anomaly) samples which are outside of the circle but close to the decision boundary (margin). This term pushes positive (anomaly) samples further away from the center c. We also add the parameter α to account for an imbalanced dataset. The loss function is then

$$Loss = \frac{1}{N}\sum_{i=1}^{N}(1-y_i)*d_i + \frac{1}{M}\sum_{j=1}^{M}\alpha * y_j * (max(0, m-d_j) + \frac{1}{1+e^{d_j-m}}) . \quad (1)$$

where N is the number of normal samples in one batch; M is the number of anomaly samples in the same batch; m is the margin, which can also be regarded as the distance from the center c to the decision boundary, and y is the label. Positive (anomaly) samples are labeled y = 1, whereas y = 0 is the label for negative (normal) samples.

The values of the different loss terms in Eq. (1) depending on the distance from the center are plotted in Fig. 3.

A center c is also employed in the deep contrastive metric learning methods Deep SVDD (Ruff et al., 2018) and Deep SAD (Ruff et al., 2019). However, these methods use an autoencoder which is trained first and then, a center c is calculated based on the average of latent representations of all normal samples in the training data. Deep SAD demonstrated fixing the center c during training can achieve smoother and faster convergence (Ruff et al., 2019). In our research, we theoretically and experimentally prove (see Appendix B) that our loss hyper-parameter center c can be chosen randomly at initialization time, and then be fixed during training.

The cDCM loss function relies on certain assumptions in order to work effectively. First, the dataset can be imbalanced, as long as the majority of samples are normal and the minority are anomalies. Additionally, the normal samples should share some similarities, although the anomaly samples can be heterogeneous. Moreover, the cDCM loss function is relatively robust for out-of-distribution anomaly samples, which means that the number of anomaly samples can be small.

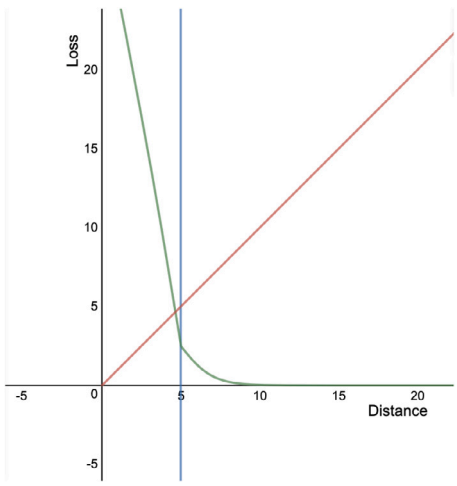


Fig. 3. Loss value plot. The red line in the plot represents the loss of samples which are labeled as negative (normal), whereas the green line represents the loss of samples which are labeled as positive (anomaly). In addition, the blue line is the decision boundary (margin).

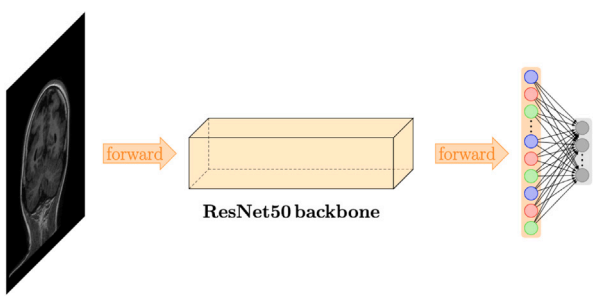


Fig. 4. Model structure for PPMR dataset classification.

Our proposed method learns a deep learning model capable of clustering normal samples together within a hypersphere while simultaneously pushing anomaly samples far away from the fixed center of the cluster. During testing, the model can simply compare the distance of the representation of a testing sample to the center of the hypersphere and then utilize the predefined decision threshold to determine whether the sample is normal or an anomaly. As a result, seen and unseen anomaly samples present dissimilar representations to normal training samples, so they are classified as outliers of the hypersphere and therefore identified as anomalies.

3.2. Model structure

In Fig. 4, we present the deep learning structure utilized for polymicrogyria prediction, where ResNet50 is utilized as the CNN feature extraction backbone. In addition, a Multi-Layer Perceptron (MLP) head maps features into a lower dimensional vector because we want to calculate an Euclidean distance metric in our novel cDCM loss function. Specifically, the MLP head has two fully connected layers with 512 and 128 neurons, respectively. Xia et al. (2015) demonstrated Euclidean distance becomes less effective with increasing dimensionality. We also note that a low dimensional latent space helps to address the "Curse of Dimensionality" when calculating distances (Aggarwal et al., 2001).

4. Datasets and experimental settings

In this section, we introduce two datasets: the pediatric polymicrogyria MRI (PPMR) dataset and the modified CIFAR10 dataset, clarify evaluation metrics, and give the experimental setting for these two datasets.

Table 1
Imaging parameters for the coronal T1 weighted sequence.

Manufacturer magnet	Siemens 3T Skyra	General electric 1.5T Cigna
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320 ∗ 260	512 ∗ 512
Slice thickness	1.2 mm	1.2 mm
FOV	20 ∗ 23 cm	22 ∗ 27 cm

4.1. Pediatric polymicrogyria MRI (PPMR) dataset

Patients under the age of 18 years with a diagnosis of PMG were identified by first searching the hospital radiology information system for polymicrogyria, then, the hospital picture archiving and communication system (PACS) was queried for MRI studies performed on these patients. The research protocol was approved by the Research Ethics Board of the Children's Hospital of Eastern Ontario. Informed consent was waived given the retrospective nature of the study.

The studies were reviewed by a fellowship trained pediatric neuroradiologist with 12 years of experience in reading pediatric brain MRI. If the presence of polymicrogyria was confirmed, the patient was included in the study. Images of the coronal 3D gradient echo T1 weighted sequence, which is the highest resolution sequence and best at showing cortical detail, were exported from the PACS system as JPEG images. One examination was chosen per patient. If there was more than one study, the most recent study was chosen. The imaging parameters for the coronal T1 weighted sequence are shown in Table 1.

For each patient, three controls were chosen and matched for gender and age (within 6 months). As for the patients, the coronal 3D gradient echo T1 weighted sequence was chosen. The images anterior to the mid-coronal plane of the orbital globes and posterior to the torcula herophili were discarded as they did not adequately show gray and white matter. The patient images were again reviewed by the radiologist and labeled as to the presence or absence of PMG. The patient cohort consists of 50% females with a mean age of 12.3 years at the time of the scan (standard deviation is 3.6 years, ranging from 5.5 years to 17.8 years). The control group is comprised of patients who had undergone MRI scans for headaches, and their MRIs were interpreted as normal. Patients with specific diagnoses of tumor, encephalitis, vascular malformation, heterotopia, focal cortical dysplasia, or phakomatoses were excluded from the control group. The control MRIs were reviewed by a pediatric radiologist (N.A.) to ensure that there were no focal or diffuse abnormalities present.

This dataset contains 23 patients in total. Within each patients MRI, not all images show evidence of PMG and some slices are normal. Although the ratio between controls and patients is 3:1, the ratio between normal slices and anomaly slices is around 5:1. Each patient's brain includes around 150 scans on average. The numbers for each patient are shown in Appendix A.

We adapt a 5-fold double cross-validation for the PPMR dataset and split these 23 patient brains into training, inner fold validation, and outer fold validation data, with a ratio of 15:4:4 at the patient level. We regard PMG scans in the inner fold validation data as '*seen anomaly*'. PMG scans in outer fold validation data are considered '*unseen anomaly*'.

To avoid data leakage, all PMG scans of one certain patient must be only in training, inner fold validation, or outer fold validation data exclusively (Yagis et al., 2019; Alinsaif et al., 2021). The patient-based data splitting method makes working with the small dataset more challenging but is crucial because of the similarity between neighboring slices for one patient.

Table 2
Data splitting example.

Data splitting	Training	Validation	Testing
Normal data amount	4000	1000	1000
Anomaly data amount	400	100	9000
Normal class ID	8	8	8
Anomaly class IDs	9,0,1,2,3	9,0,1,2,3	9,0,1,2,3,4,5,6,7

4.2. Modified CIFAR-10 dataset

We provide a modified CIFAR-10 dataset setting to mimic the challenges of the PPMR dataset. We use a modified CIFAR-10 dataset to perform exhaustive experiments, comparisons, and analysis.

The CIFAR-10 dataset is a collection of color images with the resolution of 32×32 grouped into ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship and truck. Each class contains 5000 training images and 1000 testing images. We pick one class as normal, and the rest of the 9 classes as anomalies, but only 5 anomaly classes appear in training and validation data. We call these classes that appear in training and validation data as *seen anomaly*. On the other hand, classes that only appear in testing data are called *unseen anomaly*.

Since our PPMR dataset is imbalanced and lacks data diversity, we modify the CIFAR-10 dataset to make it an imbalanced dataset. The imbalanced ratio in the modified CIFAR-10 dataset should be slightly different from the ratio between normal and anomalous samples in our PPMR dataset, which is 5:1. In our PPMR dataset, the diversity of anomalous samples is small because there are multiple consecutive MRIs which are similar to each other. In addition, the diversity of anomalous samples in the modified CIFAR-10 dataset ought to be larger than the anomalous diversity in the PPMR dataset if they have the same imbalanced ratio. Hence, we set the imbalance ratio as 10:1 of normal to anomalous samples for the modified CIFAR-10 dataset.

After modification, the amount of normal training data is $5000 * 80\% = 4000$ and the amount of anomalous training samples is $4000 * 10\% = 400$. The amount of normal validation data is $5000 * 20\% = 1000$ and the amount of anomaly validation data is $1000 * 10\% = 100$. In the testing data, we want to observe the distribution of prediction values on the whole, so we have not removed any data samples from testing data. Hence, the amount of normal testing data is 1000; and the amount of anomaly testing data is $1000 * 9 = 9000$. One detailed example where the ship class with class ID 8 is regarded as the normal class is shown in Table 2. In this case, the normal class has ID 8, seen anomaly classes have IDs 9,0,1,2 or 3 and unseen anomaly class have IDs 4,5,6 or 7.

4.3. Metric and evaluation

Accuracy, precision, recall, F_β measure, area under the curve of ROC are commonly used to evaluate the performance of binary classification tasks. The recall, also known as sensitivity, or true positive rate, is regarded as the most important metric in medical image classification (Hicks et al., 2022). The goal for our model is to miss as few positive (anomaly) samples as possible. It is unreasonable to only take recall into account because it is trivial for a classifier to attain 100% recall by simply predicting all samples as positive. Hence, the F_2 measure is chosen as our main evaluation metric. We will also report other metrics such as accuracy, area under receiver operating characteristic (ROC) curve (Mandrekar, 2010), etc.

4.4. Experimental settings on the modified CIFAR-10 dataset

Our method uses an end-to-end training strategy. When training the model using the cDCM loss function, we adopted the same LeNet-type convolutional neural network (CNN) architecture as used by Ruff et al. in their Deep SAD experiments. Specifically, we used a kernel size of 5 to train the model on the modified CIFAR-10 dataset. More hyper-parameters settings are shown in Table 3.

Table 3
Hyper-parameters setting for the modified CIFAR-10 dataset.

Hyper-parameters	Value
Margin	5
Theta	5
Latent representation dimension	128
Leaky ReLU ratio	0.01
Optimizer	Adam
Initial learning rate	1e−3
Batch size	128
Maximum training epochs	400

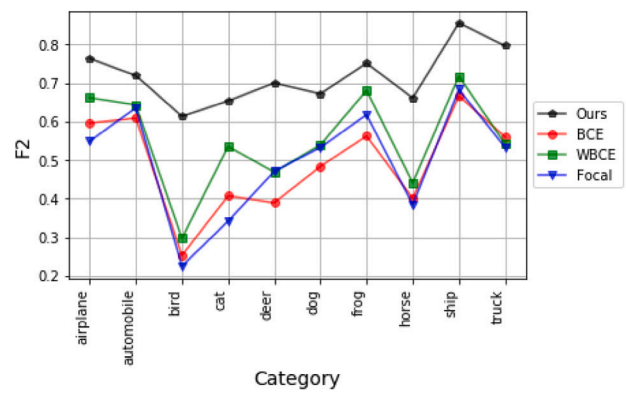


Fig. 5. Comparison of F_2 measure results between our loss function and three cross-entropy-based loss functions on modified CIFAR-10 dataset at the pre-defined decision thresholds.

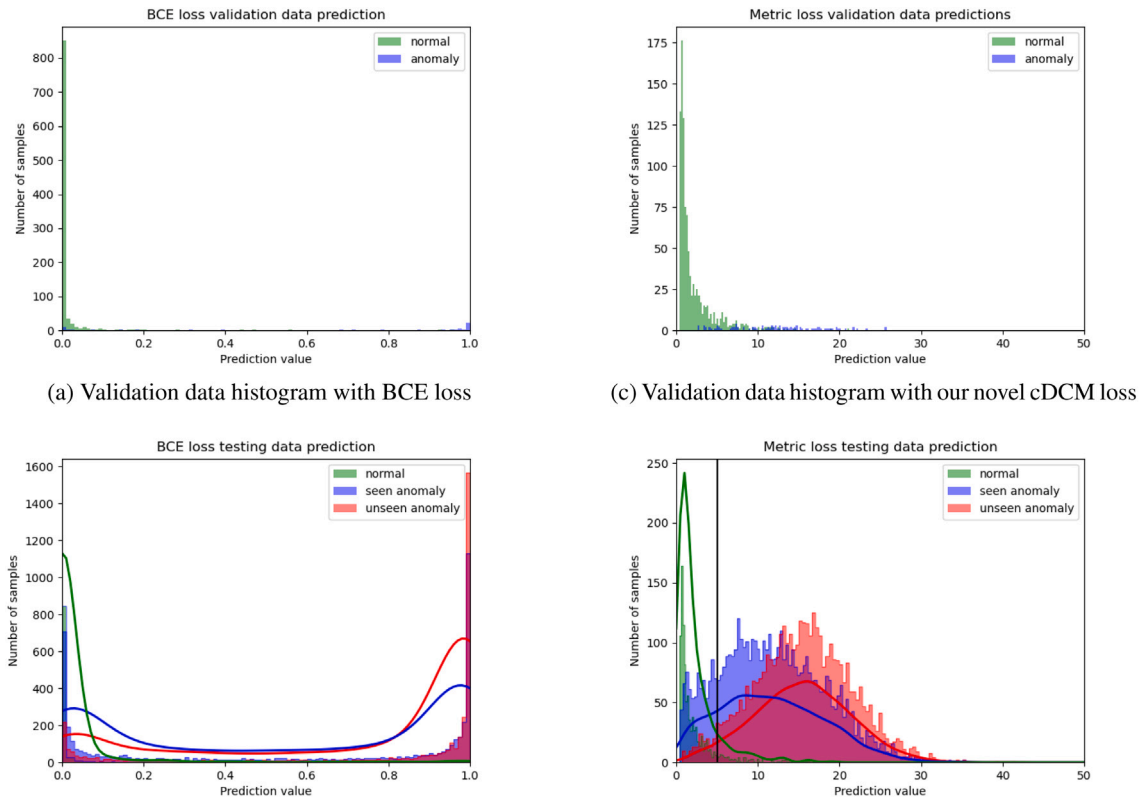
For the cross-entropy-based loss during training, we add one more fully-connected dense layer to transform the 128-dimensional vector into one probability logit value with the help of a sigmoid activation.

Since modified CIFAR-10 training and validation data lack diversity, training models is easy for a model to overfit to the training data. To relieve overfitting, we choose four different data augmentation methods (i.e., rotate, shift, shear, and zoom) during training. We rely on the AUCROC metric when minimizing the loss during the evaluation of the corresponding model parameters on the validation set. We opt not to use the F_2 measure for model selection due to potential differences in the data distribution between validation and testing datasets. Instead, we employ the AUCROC metric, as it is well-suited for evaluating a model's overall discrimination ability. If using the F_2 measure for model selection, it cannot differentiate between misclassified anomaly samples located closer to the center and other misclassified anomaly samples closer to the threshold. To better select a model that is able to distinguish samples like these, we rely on the AUCROC metric for model selection. Therefore, we save the model parameters with the best metric on the validation data during training. After inference, we find we can get a high recall and acceptable precision.

During training, we decay the learning rate with a ratio of 0.5 when validation loss is on a plateau (no decrease) within 900 iterations. We set the number of training epochs to 400 and apply early stopping when the validation loss does not decrease within 4500 iterations. To obtain more stable results, we train each model 5 times with random weight initialization and then average the results of these 5 experiments. We choose the Xavier uniform initializer (default initializer for convolution in TensorFlow). Models for the modified CIFAR-10 dataset are implemented in TensorFlow 2.8 and trained on Google Colab Pro+.

4.5. Experimental settings on our pediatric polymicrogyria MRI (PPMR) dataset

The input 2D image shape in our PPMR dataset is $256 \times 256 \times 3$, and these images are normalized to the range from 0 to 1.



(a) Validation data histogram with BCE loss

(c) Validation data histogram with our novel cDCM loss

(b) Testing data histogram with BCE loss

(d) Testing data histogram with our novel cDCM loss

Fig. 6. Prediction histograms on the modified CIFAR-10 data samples with BCE loss and our novel cDCM loss. Here, we use the class 8 (ship) as normal.

Some work (Shree and Kumar, 2018; Sathi and Islam, 2020) used skull stripping as one pre-processing step on the brain tumor challenges. But we do not choose this pre-processing step because it may lose some essential information of PMG, such as parts of gray matters close to the brain skull.

Models on our PPMR dataset are implemented in TensorFlow 2.4 and trained on a single NVIDIA GeForce RTX 3090 24 GB GPU. We use a batch size of 64 in the PPMR dataset rather than 128 as in the modified CIFAR-10 dataset to fit the model and data into memory. In addition, since we theoretically prove that our loss hyper-parameter center c can be chosen randomly at initialization (see Appendix B), we chose an arbitrary value, i.e., all 1.0, to serve as the center representation for training the PPMR dataset. We also experimentally verify that the random initialization has no impact on the modified CIFAR-10 dataset. The remaining hyper-parameters settings are the same as in Table 3 and hence the training strategies are the same as for training on the modified CIFAR-10 dataset.

Different from averaging five randomly initialized training results as for the modified CIFAR-10 dataset, we utilize 5-fold double cross-validation in our PPMR dataset. We utilize four inner folds with different validation data for each outer fold. Since our whole PPMR dataset is relatively small, validation data may be biased or be not representative. In addition, the poor quality of validation data will affect model generalization. If we want to obtain a model with good generalization, we have to save model weights where validation performance is at least good enough. Therefore, in each outer fold inference phase, we pick the model with the best validation AUCROC value in its corresponding 4 inner folds.

5. Experimental results, analysis and discussion

In this section, we show comprehensive experimental results. To explore the advantages of our cDCM loss function, we perform experiments to compare our cDCM loss function with cross-entropy-based loss functions and with the Deep SAD loss function on the modified CIFAR-10 dataset in Section 5.1 and in Section 5.2, respectively. Furthermore, Section 5.3 presents a comparison between the cDCM loss function and the triplet loss function. Our novel cDCM loss function is inspired by the Deep SAD loss function. We show that our novel cDCM loss function surpasses the state of the art. We then apply our cDCM loss function with the ResNet50 backbone to the PPMR dataset.

5.1. Comparison with cross-entropy-based loss functions

We compare the modified CIFAR-10 dataset performance of our cDCM loss and several cross-entropy-based loss functions, i.e., binary cross-entropy loss (BCE), weighted binary cross-entropy loss (WBCE), and focal loss (Lin et al., 2017). The weighted binary cross entropy loss puts more weight on the minority class in the loss function. Focal loss emphasizes learning from hard samples, including misclassified samples and minority class samples (Lin et al., 2017).

In the modified CIFAR-10 dataset, the distribution of the training and the testing data are not the same because of the unseen anomalies during training. Consequently, we choose the default threshold 0.5 for BCE, WBCE, and focal losses; and choose the hyper-parameter margin as the decision threshold for our cDCM loss. Comparative results can be found in Table 4. We can see AUCROC values of our novel cDCM loss and three cross-entropy-based loss functions are similar, but our cDCM loss function has an advantage at the predefined decision thresholds. In Fig. 5, we can see that our cDCM loss achieves better F_2 measure values in all classes, compared with other loss functions. Our novel cDCM loss function tends to provide a higher recall and only slightly lower precision.

We analyze prediction distributions on modified CIFAR-10 training, validation, and testing data separately. Results are shown in Fig. 6. Fig. 6(b) presents the testing data prediction distribution based on

Table 4
Comparison between our novel cDCM loss function, three cross-entropy-based loss functions, and the Deep SAD loss function on the modified CIFAR-10 dataset. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Loss	Precision	Recall	F_2 measure	Accuracy	AUCROC
airplane	Ours	0.9841 ± 0.0048	**0.7239** ± 0.0594	**0.7637** ± 0.0530	0.7407 ± 0.0496	0.8929 ± 0.0058
airplane	BCE	0.9929 ± 0.0010	0.5423 ± 0.0297	0.5962 ± 0.0286	0.5845 ± 0.0263	0.8870 ± 0.0034
airplane	WBCE	0.9919 ± 0.0031	0.6127 ± 0.0831	0.6622 ± 0.0772	0.6467 ± 0.0725	0.8935 ± 0.0100
airplane	Focal	0.9937 ± 0.0011	0.4956 ± 0.0941	0.5492 ± 0.0931	0.5431 ± 0.0837	0.8838 ± 0.0049
airplane	Deep SAD	0.9836 ± 0.0011	0.7128 ± 0.0236	0.7543 ± 0.0211	0.7308 ± 0.0203	0.8840 ± 0.0060
automobile	Ours	0.9934 ± 0.0011	0.6735 ± 0.0172	0.7198 ± 0.0157	0.7021 ± 0.0150	0.8927 ± 0.0050
automobile	BCE	0.9978 ± 0.0003	0.5552 ± 0.0262	0.6091 ± 0.0253	0.5985 ± 0.0234	0.9162 ± 0.0054
automobile	WBCE	0.9972 ± 0.0012	0.5915 ± 0.0388	0.6436 ± 0.0364	0.6308 ± 0.0342	0.9197 ± 0.0037
automobile	Focal	0.9965 ± 0.0012	0.5832 ± 0.0279	0.6358 ± 0.0265	0.6230 ± 0.0244	0.9153 ± 0.0027
automobile	Deep SAD	0.9876 ± 0.0013	**0.7302** ± 0.0159	**0.7703** ± 0.0141	0.7489 ± 0.0138	0.8744 ± 0.0132
bird	Ours	0.9576 ± 0.0091	**0.5641** ± 0.0552	**0.6140** ± 0.0518	0.5849 ± 0.0436	0.7310 ± 0.0190
bird	BCE	0.9827 ± 0.0049	0.2139 ± 0.0465	0.2531 ± 0.0516	0.2890 ± 0.0405	0.7363 ± 0.0145
bird	WBCE	0.9815 ± 0.0022	0.2534 ± 0.0204	0.2975 ± 0.0225	0.3238 ± 0.0179	0.7483 ± 0.0068
bird	Focal	0.9805 ± 0.0052	0.1896 ± 0.0499	0.2255 ± 0.0562	0.2671 ± 0.0434	0.7257 ± 0.0107
bird	Deep SAD	0.9667 ± 0.0013	0.3246 ± 0.0611	0.3739 ± 0.0646	0.3823 ± 0.0532	0.7197 ± 0.0111
cat	Ours	0.9662 ± 0.0038	**0.6053** ± 0.0479	**0.6538** ± 0.0448	0.6256 ± 0.0396	0.7741 ± 0.0056
cat	BCE	0.9813 ± 0.0057	0.3566 ± 0.0765	0.4074 ± 0.0811	0.4145 ± 0.0659	0.7679 ± 0.0096
cat	WBCE	0.9717 ± 0.0155	0.4881 ± 0.1877	0.5354 ± 0.1787	0.5243 ± 0.1552	0.7749 ± 0.0186
cat	Focal	0.9825 ± 0.0031	0.2944 ± 0.0427	0.3419 ± 0.0459	0.3601 ± 0.0370	0.7548 ± 0.0072
cat	Deep SAD	0.9750 ± 0.0020	0.4769 ± 0.1082	0.5290 ± 0.1104	0.5181 ± 0.0944	0.7609 ± 0.0237
deer	Ours	0.9728 ± 0.0018	**0.6546** ± 0.0418	**0.7002** ± 0.0385	0.6727 ± 0.0358	0.8261 ± 0.0058
deer	BCE	0.9927 ± 0.0024	0.3383 ± 0.0255	0.3896 ± 0.0271	0.4022 ± 0.0227	0.8251 ± 0.0059
deer	WBCE	0.9873 ± 0.0105	0.4209 ± 0.1828	0.4692 ± 0.1768	0.4725 ± 0.1561	0.8267 ± 0.0114
deer	Focal	0.9873 ± 0.0126	0.4224 ± 0.1726	0.4712 ± 0.1681	0.4736 ± 0.1461	0.8189 ± 0.0094
deer	Deep SAD	0.9810 ± 0.0028	0.5420 ± 0.0599	0.5947 ± 0.0578	0.5782 ± 0.0515	0.8272 ± 0.0072
dog	Ours	0.9852 ± 0.0033	**0.6231** ± 0.0402	**0.6722** ± 0.0372	0.6522 ± 0.0337	0.8624 ± 0.0043
dog	BCE	0.9936 ± 0.0026	0.4293 ± 0.0412	0.4839 ± 0.0421	0.4838 ± 0.0359	0.8387 ± 0.0054
dog	WBCE	0.9902 ± 0.0051	0.4856 ± 0.0734	0.5397 ± 0.0715	0.5324 ± 0.0629	0.8439 ± 0.0058
dog	Focal	0.9885 ± 0.0100	0.4815 ± 0.1358	0.5331 ± 0.1301	0.5273 ± 0.1151	0.8389 ± 0.0029
dog	Deep SAD	0.9849 ± 0.0013	0.6172 ± 0.0139	0.6670 ± 0.0129	0.6469 ± 0.0118	0.8454 ± 0.0097
frog	Ours	0.9909 ± 0.0016	**0.7090** ± 0.0142	**0.7518** ± 0.0126	0.7323 ± 0.0117	0.9107 ± 0.0056
frog	BCE	0.9985 ± 0.0011	0.5077 ± 0.0357	0.5628 ± 0.0349	0.5561 ± 0.0316	0.9066 ± 0.0067
frog	WBCE	0.9943 ± 0.0043	0.6329 ± 0.0969	0.6810 ± 0.0896	0.6880 ± 0.0924	0.9139 ± 0.0042
frog	Focal	0.9968 ± 0.0020	0.5651 ± 0.0608	0.6180 ± 0.0587	0.6069 ± 0.0537	0.9154 ± 0.0052
frog	Deep SAD	0.9904 ± 0.0013	0.7047 ± 0.0180	0.7478 ± 0.0163	0.7281 ± 0.0158	0.9027 ± 0.0113
horse	Ours	0.9847 ± 0.0058	**0.6120** ± 0.0689	**0.6613** ± 0.0647	0.6420 ± 0.0583	0.8510 ± 0.0172
horse	BCE	0.9982 ± 0.0008	0.3493 ± 0.0399	0.4012 ± 0.0419	0.4138 ± 0.0356	0.8807 ± 0.0075
horse	WBCE	0.9972 ± 0.0016	0.3883 ± 0.0712	0.4414 ± 0.0729	0.4484 ± 0.0633	0.8852 ± 0.0029
horse	Focal	0.9981 ± 0.0004	0.3332 ± 0.0186	0.3844 ± 0.0198	0.3993 ± 0.0167	0.8854 ± 0.0050
horse	Deep SAD	0.9844 ± 0.0021	0.5916 ± 0.0388	0.6427 ± 0.0367	0.6240 ± 0.0338	0.8236 ± 0.0193
ship	Ours	0.9880 ± 0.0028	0.8281 ± 0.0331	0.8557 ± 0.0280	0.8362 ± 0.0279	0.9412 ± 0.0062
ship	BCE	0.9954 ± 0.0014	0.6166 ± 0.0733	0.6665 ± 0.0698	0.6523 ± 0.0650	0.9373 ± 0.0039
ship	WBCE	0.9940 ± 0.0009	0.6714 ± 0.0179	0.7179 ± 0.0163	0.7006 ± 0.0155	0.9395 ± 0.0023
ship	Focal	0.9938 ± 0.0015	0.6347 ± 0.0778	0.6831 ± 0.0730	0.6675 ± 0.0688	0.9339 ± 0.0023
ship	Deep SAD	0.9835 ± 0.0023	**0.8411** ± 0.0231	**0.8661** ± 0.0194	0.8443 ± 0.0195	0.9265 ± 0.0092
truck	Ours	0.9828 ± 0.0020	**0.7610** ± 0.0298	**0.7968** ± 0.0260	0.7729 ± 0.0250	0.8974 ± 0.0076
truck	BCE	0.9955 ± 0.0009	0.5057 ± 0.0494	0.5604 ± 0.0489	0.5530 ± 0.0439	0.9006 ± 0.0073
truck	WBCE	0.9963 ± 0.0008	0.4874 ± 0.0378	0.5426 ± 0.0375	0.5370 ± 0.0336	0.9045 ± 0.0030
truck	Focal	0.9954 ± 0.0025	0.4793 ± 0.0764	0.5337 ± 0.0756	0.5292 ± 0.0673	0.9032 ± 0.0065
truck	Deep SAD	0.9824 ± 0.0034	0.7521 ± 0.0154	0.7891 ± 0.0133	0.7648 ± 0.0122	0.8769 ± 0.0092

the BCE loss. Although more than half of anomaly samples are predicted correctly as anomalous, a large number of anomaly samples are predicted as normal. We can see that this BCE loss model seems to overfit the data; because the predictions on the validation data are perfect (see Fig. 6(a)), but the predictions on the testing data are poor. However, using the resulting model obtained with our novel cDCM loss function results in a distribution of testing data predictions in the approximate shape of a normal distribution (see Fig. 6(d)). Moreover, based on Fig. 6(b), simply moving the decision threshold for the BCE loss cannot trade-off precision and recall easily because the model trained on the BCE loss fails to recognize too many anomalous samples with high confidence. For example, if one prefers recall to precision, it is reasonable to choose a low threshold $\ll 0.5$ based on Fig. 6(a). However, it does not improve recall too much on the testing data which can be seen from the tests in Appendix D.

Fig. 6(d) represents the testing data prediction distribution based on our novel cDCM loss model. The black vertical line at a value of 5 on the horizontal axis represents the margin (or decision boundary). We can see only a small part of the samples are predicted incorrectly. Moreover, the majority of normal samples are predicted close to the center c, and the majority of anomaly samples are predicted outside of the decision boundary. In our PPMR dataset, we do not have enough testing data to represent the whole distribution of anomaly samples, and our PPMR testing data are sampled from the whole distribution of anomaly samples. Based on Fig. 6(d), if testing data are randomly sampled from the whole distribution, recall remains still high, but precision will decrease dramatically. This is the main reason why our PPMR dataset tends to result in high recall and high AUCROC, but relatively low precision results when we directly use the predefined margin as the decision threshold. However, precision and recall can

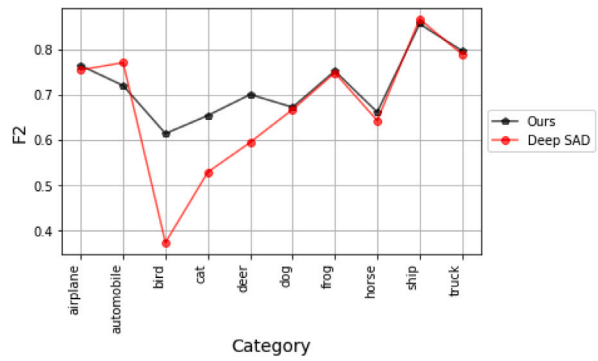


Fig. 7. Comparison of F_2 measure results between our loss function and Deep SAD loss on modified CIFAR-10 dataset.

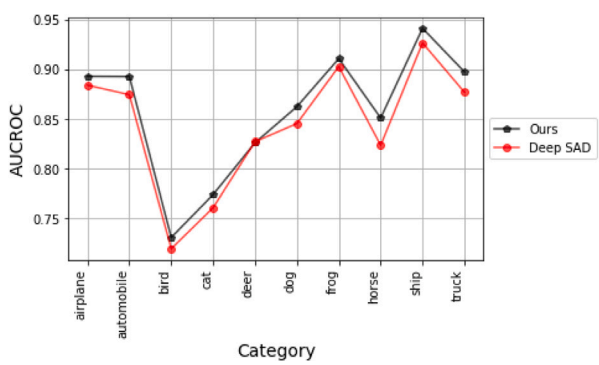


Fig. 8. Comparison of AUCROC results between our loss function and Deep SAD loss on modified CIFAR-10 dataset.

be simply balanced by moving the threshold with our novel cDCM loss function, although we prefer recall over precision in our application.

5.2. Comparison with deep SAD

Both the Deep SAD loss and our novel cDCM loss are center-based loss functions. However, the decision threshold for the Deep SAD loss is not defined. Zhou et al. (2021) use 95% percentile of distance values from all training normal samples to the center as the decision threshold. For this reason, we also choose this method to calculate the decision threshold for Deep SAD.

Comparisons of the modified CIFAR-10 dataset are also listed in Table 4. We can see recall values, F_2 measure values, accuracy values, and AUCROC values of our novel cDCM loss are better than the performance of Deep SAD loss in the majority of classes, although there is a small precision decrease. These results show the advantages of our novel cDCM loss function, compared to the Deep SAD loss function. This is probably because our cDCM loss function has a clear predefined decision threshold which can distinguish confusing samples but the Deep SAD loss function may not disentangle these confusing samples.

Fig. 7 shows the comparison of F_2 measure results. In the majority of categories, our novel cDCM loss function performs better or slightly better, with the difference sometimes being less than 1%, than the Deep SAD loss, except for the category of automobile and ship. In addition, we can see there is a huge performance improvement by more than 10% when the category bird, cat, or deer is the normal class making our model performance more equal between classes. Hence, our novel cDCM loss function is more stable than the Deep SAD loss. Fig. 8 shows the comparison of AUCROC results. We can see the AUCROC values of our novel cDCM loss function are higher than the AUCROC values of the Deep SAD loss function in all categories.

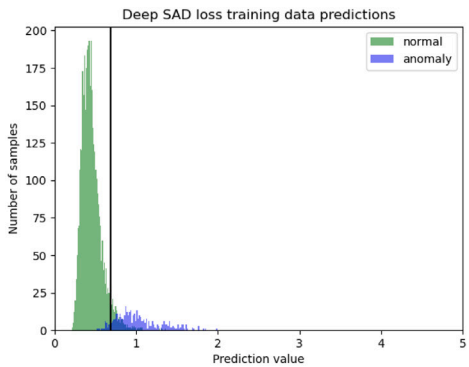
In Table 4 and Fig. 7, we can see when the class 2 (bird) is the normal class, the AUCROC scores of our loss function and Deep SAD loss function are similar (0.731 vs. 0.7197) but the F_2 measure values of these two loss functions have a large difference (0.6140 vs. 0.3739). To analyze this phenomenon, prediction distribution plots are shown in Fig. 9. Since the results presented in Table 4 represent the average of five experiments, each with different weight initialization, drawing the prediction distribution only requires one experiment. Thus, we select one representative experiment. In this particular experiment, the AUCROC score for cDCM loss function stands at 0.7375, while Deep SAD loss function achieves the AUCROC score of 0.7032. Furthermore, the F_2 measure values of these two loss functions are 0.6986 and 0.337, respectively. In Fig. 9(b), the black vertical line is the decision threshold, which is the 95% percentile of distance values from all training normal samples to the center. We can see distributions of normal, seen anomaly, and unseen anomaly are close to each other and the majority of values of these three distributions are all on the left side of the decision threshold. Moreover, distributions of normal and unseen anomaly are almost the same, which is not beneficial in distinguishing them. Different from the distributions with the Deep SAD loss, the distributions with our novel cDCM loss are wide and the majority of values of the seen anomaly sample distribution are on the right of the decision boundary, while the majority of values of the normal sample distribution are on the left side of the decision boundary. In this case, the normal class is birds, while the seen anomaly classes are all animals, including cats, deer, dogs, frogs, and horses. However, the unseen anomaly classes are all vehicles, such as ships, trucks, airplanes, and automobiles. Despite not appearing in the training and validation data, our novel cDCM loss function can still distinguish these unseen vehicles as anomalies.

Overall, there is probably no good decision threshold for the Deep SAD loss function, but we can define a decision threshold for our novel cDCM loss function first, and the deep learning model will optimize our novel cDCM loss function and separate normal and anomaly distributions into the two sides of the decision threshold. In our method a good decision threshold can be chosen for imbalanced datasets where the distribution of the anomaly samples is only partially known. This is the main reason why our novel cDCM loss can achieve better F_2 measure than the Deep SAD loss, although they have similar AUCROC scores.

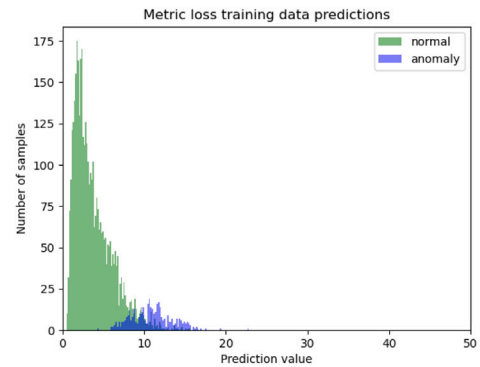
Moreover, we perform experiments on the modified CIFAR-10 dataset in Appendix B.2 to verify that choosing different centers in our novel cDCM loss function does not affect the final performance too much. Additionally, choosing different margins in our novel cDCM loss function also does not affect the final performance too much and these results are given in Appendix B.3. In addition, we conduct the tests of statistical significance between 5 different loss functions on the modified CIFAR-10 dataset in Appendix C.

5.3. Comparison with triplet loss functions

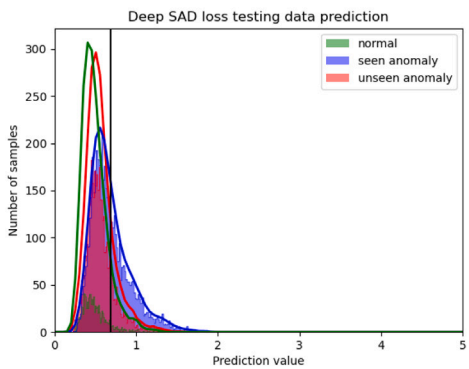
Triplet loss is also commonly used in supervised anomaly detection (Ju et al., 2020; Du and Zhang, 2014). Ju et al. (2020) introduced a method called "PUMAD" that was originally designed for an unlabeled dataset with a few labeled anomaly samples. We adapted this method to our evaluation by constructing triplets from labeled normal and anomaly samples. During inference, the anomaly score is calculated as the average distance from a target sample to all normal training data samples, which is used to compute the AUCROC score. However, this method does not use a decision threshold, so we only report the AUCROC score for PUMAD. One drawback of the PUMAD method is its high time complexity during inference. Specifically, when the number of testing samples is T and the number of normal training samples is N, the Euclidean distance needs to be calculated $T \times N$ times. This can quickly become prohibitively expensive as the number of normal training samples grows. However, the learned representations from the triplet loss can be applied to a classifier, such as a support vector



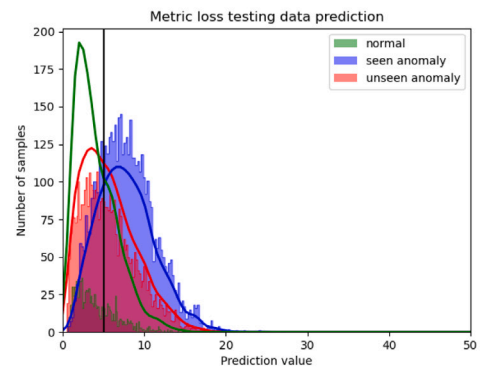
(a) Training data histogram with Deep SAD loss



(c) Training data histogram with our novel cDCM loss



(b) Testing data histogram with Deep SAD loss



(d) Testing data histogram with our novel cDCM loss

Fig. 9. Modified CIFAR-10 data samples prediction distribution with the Deep SAD loss and with our novel cDCM loss. The class 2 (bird) is the normal class. Please note: the x-axis is range from 0 to 5 in the left three plots, while the x-axis is range from 0 to 50 in the right three plots. The black vertical line is the decision threshold.

Table 5
Comparison between our novel cDCM loss function, PUMAD, and Triplet loss function with one SVM classifier on the modified CIFAR-10 dataset. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Method	Precision	Recall	F_2 measure	Accuracy	AUCROC
airplane	cDCM	0.9841 ± 0.0048	0.7239 ± 0.0594	**0.7637** ± 0.0530	0.7407 ± 0.0496	0.8929 ± 0.0058
airplane	PUMAD					0.8908 ± 0.0047
airplane	Triplet+SVM	0.9953 ± 0.0012	0.4045 ± 0.0057	0.4590 ± 0.0058	0.4623 ± 0.0046	0.6937 ± 0.0010
automobile	cDCM	0.9934 ± 0.0011	0.6735 ± 0.0172	**0.7198** ± 0.0157	0.7021 ± 0.0150	0.8927 ± 0.0050
automobile	PUMAD					0.8951 ± 0.0033
automobile	Triplet+SVM	0.9974 ± 0.0004	0.5384 ± 0.0215	0.5929 ± 0.0208	0.5833 ± 0.0195	0.7632 ± 0.0116
bird	cDCM	0.9576 ± 0.0091	0.5641 ± 0.0552	**0.6140** ± 0.0518	0.5849 ± 0.0436	0.7310 ± 0.0190
bird	PUMAD					0.6964 ± 0.0141
bird	Triplet+SVM	0.9842 ± 0.0016	0.1170 ± 0.0083	0.1421 ± 0.0098	0.2036 ± 0.0074	0.5501 ± 0.0038
cat	cDCM	0.9662 ± 0.0038	0.6053 ± 0.0479	**0.6538** ± 0.0448	0.6256 ± 0.0396	0.7741 ± 0.0056
cat	PUMAD					0.7511 ± 0.0175
cat	Triplet+SVM	0.9901 ± 0.0023	0.1935 ± 0.0309	0.2304 ± 0.0352	0.2723 ± 0.0272	0.5878 ± 0.0123
deer	cDCM	0.9728 ± 0.0018	0.6546 ± 0.0418	**0.7002** ± 0.0385	0.6727 ± 0.0358	0.8261 ± 0.0058
deer	PUMAD					0.8337 ± 0.0110
deer	Triplet+SVM	0.9920 ± 0.0008	0.2657 ± 0.0171	0.3112 ± 0.0188	0.3372 ± 0.0154	0.6233 ± 0.0082
dog	cDCM	0.9852 ± 0.0033	0.6231 ± 0.0402	**0.6722** ± 0.0372	0.6522 ± 0.0337	0.8624 ± 0.0043
dog	PUMAD					0.8508 ± 0.0074
dog	Triplet+SVM	0.9950 ± 0.0008	0.3485 ± 0.0063	0.4005 ± 0.0067	0.4120 ± 0.0058	0.6665 ± 0.0040
frog	cDCM	0.9909 ± 0.0016	0.7090 ± 0.0142	**0.7518** ± 0.0126	0.7323 ± 0.0117	0.9107 ± 0.0056
frog	PUMAD					0.9010 ± 0.0072
frog	Triplet+SVM	0.9986 ± 0.0003	0.4186 ± 0.0115	0.4736 ± 0.0118	0.4762 ± 0.0105	0.7068 ± 0.0062
horse	cDCM	0.9847 ± 0.0058	0.6120 ± 0.0689	**0.6613** ± 0.0647	0.6420 ± 0.0583	0.8510 ± 0.0172
horse	PUMAD					0.8318 ± 0.0274
horse	Triplet+SVM	0.9972 ± 0.0004	0.3669 ± 0.0135	0.4200 ± 0.0141	0.4292 ± 0.0121	0.6789 ± 0.0067
ship	cDCM	0.9880 ± 0.0028	0.8281 ± 0.0331	**0.8557** ± 0.0280	0.8362 ± 0.0279	0.9412 ± 0.0062
ship	PUMAD					0.9315 ± 0.0065
ship	Triplet+SVM	0.9957 ± 0.0005	0.5643 ± 0.0127	0.6178 ± 0.0122	0.6056 ± 0.0113	0.7713 ± 0.0057
truck	cDCM	0.9828 ± 0.0020	0.7610 ± 0.0298	**0.7968** ± 0.0260	0.7729 ± 0.0250	0.8974 ± 0.0076
truck	PUMAD					0.8855 ± 0.0083
truck	Triplet+SVM	0.9959 ± 0.0006	0.4553 ± 0.0125	0.5107 ± 0.0125	0.5080 ± 0.0110	0.7194 ± 0.0053

Table 6
Performance comparison of several loss functions on the PPMR dataset: Average of 5-fold cross-validation results with standard deviations. ResNet50 served as the feature extraction backbone in all experiments.

Loss function	Precision	Recall	F_2 measure	Accuracy	AUCROC
cDCM loss	0.7186 ± 0.1693	**0.8807 ± 0.0333**	**0.8426 ± 0.0397**	**0.9227** ± 0.0279	0.9579 ± 0.0270
BCE loss	0.7407 ± 0.1979	0.7337 ± 0.1603	0.7351 ± 0.1667	0.9043 ± 0.0220	0.9601 ± 0.0212
Deep SAD loss	0.2973 ± 0.1050	0.6455 ± 0.2442	0.5056 ± 0.1443	0.6804 ± 0.1210	0.6954 ± 0.1862

Table 7
Model comparison: average of 5-fold cross-validation results with standard deviation.

Backbone model	Precision	Recall	F_2 measure	Accuracy	AUCROC
ResNet50 + cDCM loss	0.7186 ± 0.1693	0.8807 ± 0.0333	**0.8426 ± 0.0397**	**0.9227** ± 0.0279	0.9579 ± 0.0270
MobileNet + cDCM loss	0.3708 ± 0.2928	**0.9765** ± 0.0355	0.6366 ± 0.2321	0.4786 ± 0.4013	0.9623 ± 0.0232
EfficientNet + cDCM loss	0.5535 ± 0.3570	0.7441 ± 0.4196	0.6823 ± 0.3854	0.8950 ± 0.0640	0.9598 ± 0.0294

machine (SVM) (Qu et al., 2020; Tayeh et al., 2020). Experimental results on the modified CIFAR-10 dataset are shown in Table 5. We can see that the difference in AUCROC between our cDCM loss function and the PUMAD method is minor. However, the recall, F_2 measure, accuracy, and AUCROC of our cDCM loss function is much higher than those of the triplet loss with a SVM classifier.

5.4. Main results on the PPMR dataset

We apply our cDCM loss in the training of a model with a ResNet50 backbone on the PPMR dataset. The above experiments show that our novel cDCM loss function is able to predefine a better decision threshold than binary cross-entropy-based loss functions and the earlier Deep SAD loss on the modified CIFAR-10 dataset. Since our PPMR dataset is small, we have experimented with transfer learning of a ResNet50 backbone pretrained on ImageNet. We find this model initialized with transfer learning will eventually predict all samples as positive (anomaly). Hence, we do not use transfer learning from the ImageNet dataset for our specific challenge.

In addition to our proposed cDCM loss function, we also evaluated the performance of a binary cross-entropy (BCE) loss function and the Deep SAD loss function on the PPMR dataset. Table 6 presents the results. Using ResNet50 as the backbone, our cDCM loss function achieved the highest recall, F_2 measure, and accuracy. However, the precision and AUCROC were slightly lower than those obtained with the BCE loss. Moving the threshold can also improve recall with decreasing precision. However, it is difficult to find an expected threshold without knowing the testing data in practice. Based on our pre-defined decision boundary, our cDCM loss function can directly achieve a high recall and an acceptable precision, resulting in a good F_2 measure. Moreover, the standard deviations of recall and F_2 measure were smaller for our loss function compared to the other two, indicating that our method can produce more stable results. In addition, the Deep SAD loss function performs poorly probably because it struggles to differentiate between normal and anomaly samples when they are extremely similar. Additionally, there is no specific predefined decision threshold in Deep SAD loss function to clearly separate normal and anomaly samples. While there is no significant difference between the performance of cDCM and Deep SAD at the 0.05 threshold on the modified CIFAR-10 dataset, as shown in Appendix C, cDCM outperforms Deep SAD in all metrics on the PPMR dataset.

Initially, the ResNet50 results were found to be unstable due to the occurrence of a NAN loss during training with the cDCM loss function. We observed that the model predicted an increasing number of samples as anomaly at an increasingly large distance to the center. Once one normal sample is incorrectly predicted as an anomaly, it will leads to a NAN loss. To address this issue, we introduced distance clipping from the center with a maximum distance threshold of 20. This resolves the NAN problem and results in stable results.

In our application, we prioritize the results obtained using our proposed cDCM loss function because they are more stable and offer a higher F_2 measure compared to the other two loss functions. While the choice of loss function plays a crucial role in the model's performance, there is still potential for exploring more advanced model structures in the future to further improve the accuracy of polymicrogyria detection.

Next, based on our cDCM loss function, we evaluated three popular convolutional neural network backbones: ResNet50 (He et al., 2016), MobileNet (Howard et al., 2017), and EfficientNet (Tan and Le, 2019). The results are presented in Table 7. Our findings show that ResNet50 with cDCM loss achieves the best F_2 measure score and accuracy on the PPMR dataset. Additionally, ResNet50 exhibits the smallest standard deviation in F_2 score among the three backbones. It is worth noting that MobileNet with cDCM loss has the highest average recall at 0.9765. However, its precision is extremely low at 0.3708, making it unsuitable for practical use. This is due to the fact that MobileNet with cDCM loss tends to predict almost all samples as positive in some folds.

Finally, results of the comparison between the cDCM loss function with the ResNet50 backbone and several state-of-the-art methods, like PaDiM and PANDA, as discussed in Section 2.3, are shown in Table 8. As the authors of PaDiM and PANDA did not specify how to determine the decision threshold, we omitted the results for precision, recall, F_2 measure and accuracy for these two methods in the table. The AUCROC of our method is dramatically better than of these two methods mainly because PaDiM and PANDA rely heavily on pretrained weights from ImageNet while transfer learning from the ImageNet dataset poses a significant challenge in the context of this specific task.

Transformer (Vaswani et al., 2017) structures are also commonly used in computer vision and medical imaging domains. Variants of Transformers tailored for these fields include the Vision Transformer (ViT) (Dosovitskiy et al., 2020) and the Swin Transformer (Liu et al., 2021). However, it is worth noting that Transformer models typically demand a substantial dataset for training. While we experimented the Vision Transformer as the backbone, the model did not converge when incorporating our cDCM loss function. Specifically, when the distance clipping is applied in the cDCM loss function, the model tends to predict all samples as exceeding the maximum distance threshold for several training iterations. Conversely, if distance clipping is not applied, the model encounters a NAN loss. In either case, the model fails to learn any meaningful information due to the absence of gradients between iterations. In general, we recommend the current cDCM loss for models with fewer parameters well matched to the task at hand. For instance, when applied to a model with three basic CNN layers (similar to a LeNet-type CNN) on the modified CIFAR10 dataset, the cDCM loss function, without distance clipping, never leads to NAN loss issues. However, when using larger CNN backbones like ResNet50 with the cDCM loss function, NAN loss may occasionally occur. These issues can be mitigated by implementing distance clipping. However, when the cDCM loss function is applied with distance clipping to large Transformer backbones, which contain a larger number of parameters compared to large CNN backbones, NAN loss issues consistently arise. In the future, there is a need to explore more advanced cDCM-type loss functions that are better suited for Transformer model structures.

Table 8
Comparison of our method with the state-of-the-art for anomaly detection: average of 5-fold cross-validation results with standard deviation.

Method	AUCROC
ResNet50 + cDCM loss	**0.9579 ± 0.0270**
PaDiM	0.8524 ± 0.0432
PANDA	0.6877 ± 0.0722

6. Conclusions and future work

Inexperienced readers could misclassify polymicrogyria (PMG) MRIs as normal MRIs in practice, and a computer-aided method to detect PMG in MRIs is would be of value. In the clinical setting, it is important to identify all cases of PMG, so a high recall is important. Our model provides a high recall and an acceptable precision and is therefore promising in the future evaluation of PMG cases.

We find cross-entropy-based loss functions predict the majority of images as normal, which causes results of high precision but low recall. In addition, models with cross-entropy-based loss functions easily lead to overfitting and they lack the ability to generalize, especially, if the sample contains unseen features during training. Moreover, one cannot trade-off precision and recall by simply moving the decision threshold in cross-entropy-based loss functions because of out-of-distribution samples. However, our novel cDCM loss function tends to predict images with high F_2 measure, which means relatively high recall and acceptable precision, and partly deals with the generalization problem. Finally, a ResNet50 backbone with our novel cDCM loss function can achieve 88.07% recall at 71.86% precision on PPMR dataset.

This research has a number of shortcomings. Radiologists will find a substantial number of normal MRIs in the set of possible PMG MRIs predicted by our algorithm since the precision of our final results is 71.86%. Furthermore, because of the limited size of our PPMR dataset, it is challenging to assess how well the model will perform on the actual PMG distribution. To our knowledge, this study is the first to distinguish PMG MRIs from normal MRIs by machine learning methods and hence there is a lot of work that needs to be done in the future.

As for future work, our method is promising for some other kinds of medical imaging classification tasks, especially for some tasks where out-of-distribution samples will be encountered when deployed. However, precision needs be increased further without decreasing recall. To achieve this goal, more effective data augmentation and improving the model structure can be explored further.

CRediT authorship contribution statement

Lingfeng Zhang: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Nishard Abdeen:** Conceptualization, Data curation, Writing – review & editing. **Jochen Lang:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

None Declared.

Data availability

Data will be made available on request.

Acknowledgment

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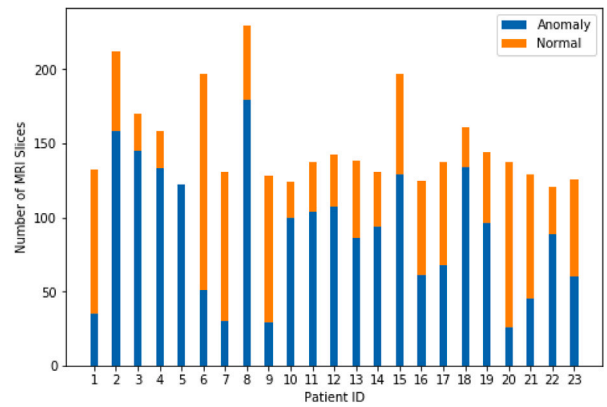

Fig. 10. The amount of normal slices and anomaly slices for each patient.

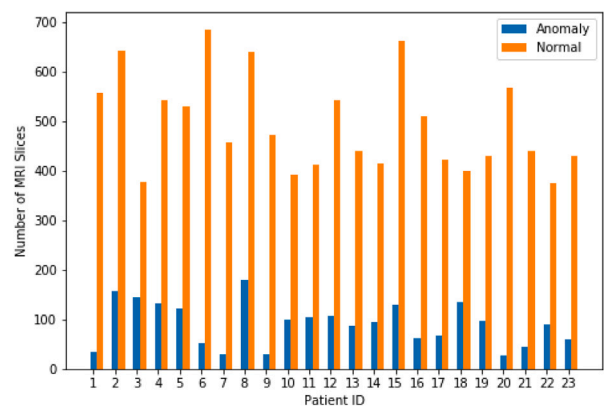

Fig. 11. Our hospital polymicrogyria dataset MRI slices distribution.

Appendix A. Distribution of the PPMR dataset

The ratio between the amount of normal MRI slices and the amount of anomaly MRI slices for each patient are shown in Fig. 10. And the whole amount of normal MRI slices and anomaly MRI slices for each patient with 3 controls are shown in Fig. 11. We can see this is an imbalanced dataset.

Appendix B. Choosing the center and margin for our cDCM loss

One of the key advantages of our proposed cDCM loss is the fact that a center in the latent space can be chosen randomly. Below we give a proof to elucidate why this is correct, followed by an experimental verification. The margin is another important hyper-parameter in our novel cDCM loss function. In Appendix B.3, we experimental show that this margin can also be chosen randomly within reason.

B.1. Theoretical considerations

Proposition. *Given a specific deep learning structure, ideally, if one center c' is initialized randomly and is fixed during training, one model trained on this center c' can achieve the same optimum as other models trained on other centers. It is not necessary to find the true center c first.*

Proof.

1. Assume there is a true center $c(c_0, c_1, c_2, \dots, c_n)$ of the latent representations of all normal samples, and c is a n + 1 dimensional vector
2. Our randomly defined center is $c'(c_0+s_0, c_1+s_1, c_2+s_2, \dots, c_n+s_n)$, where s_n is a constant value.

3. Assume there is an *hypothetical* well-trained model f based on the correct center c, so $f(x) = (l_0, l_1, l_2, \dots, l_n)$. l_i represents the ith value of the latent representation l in dimension i.
4. Our well-trained model based on our randomly initialized center c' can be represented as $f'(x) = (l'_0, l'_1, l'_2, \dots, l'_n)$.
5. The squared distance between the latent representation of a sample in the hypothetical well-trained model f from the true center is $d^2 = (l_0 - c_0)^2 + (l_1 - c_1)^2 + (l_2 - c_2)^2 + \cdots + (l_n - c_n)^2$
6. And the squared distance between the latent representation of a sample in our well-trained model f' from the randomly defined center is $d'^2 = (l'_0 - c_0 - s_0)^2 + (l'_1 - c_1 - s_1)^2 + (l'_2 - c_2 - s_2)^2 + \cdots + (l'_n - c_n - s_n)^2$
7. If there exists a simple method to ensure that $d^2 == d'^2$ for each sample, our well-trained model f' based on center c' will have the same global loss value as the well-trained model f based on the center c. An obvious solution would be

$$\begin{aligned} l_0 &== l'_0 - s_0 \\ l_1 &== l'_1 - s_1 \\ l_2 &== l'_2 - s_2 \\ &\cdots \\ l_n &== l'_n - s_n \,. \end{aligned}$$

8. We can write l'_n as $l_n + k_n$, where k_n is another constant value. If we want $l_n == l'_n - s_n$, we must ensure $l_n == l'_n + k_n - s_n$ and hence we have to set $k_n == s_n$.
9. We consider s_n as the shift of the true center c. And intuitively, k_n can be regarded as the shift of latent representations in the latent space. A shift of the latent representations can be achieved by the bias term of the last fully-connected layer in the model. Hence, although the loss is only minimized for all training samples as a whole, there exists a simple solution for deep learning models to meet $d^2 == d'^2$ for each sample by modifying only the bias term.
10. The minimum loss value is 0 given our novel cDCM loss function. In this case, all normal samples will be located at the position of the center in the latent representation.
11. Hence, shifting centers means shifting all normal samples in the same direction and it is not necessary to find the true center based on normal samples.

Overall, the training process could simply ensure that the correct center c shifts to random center c', and the latent representation of each sample based on the correct center c will also be shifted, to keep the same distance from each data sample to the center. However, if the loss does not reach 0 for the normal samples, and considering the loss caused by the anomaly samples, there could be other complex transformations for these latent representations to make sure the global loss is optimal.

In addition, different model structures likely map samples differently into the latent space and hence some may be better than others. Therefore, model structure is also important to achieve good performance.

B.2. Experimental comparison of different centers

On modified CIFAR-10 dataset, we try three different centers during our novel cDCM loss model training. The first center 128-dimensional all zero vector; the second center is all ones, and the last one is random value drawn from an uniform sampling distribution in the range between 0.0 to 1.0. In Appendix B.2, we show that the center of our novel cDCM loss can be chosen randomly under the assumption that the model is well trained. In our experiments, we also show that given a specific model structure, choosing different centers in initialization can achieve almost the same results. In Table 9, we can see performance results, especially AUCROC, for different centers are almost the same in all classes. The small difference are likely caused by weight initialization and shuffling batches during training. We save the model with the best validation AUCROC value but a small AUCROC value difference may move the decision threshold by a relatively large amount. This is the likely cause for the performance metrics, except for AUCROC, exhibit small differences in some cases with different centers. The overall results clearly demonstrate that the center can be chosen randomly in practice.

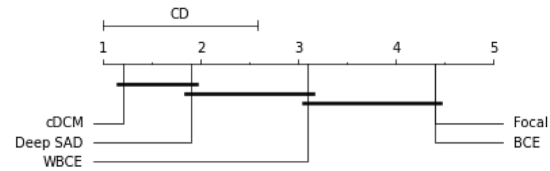

Fig. 12. Critical difference diagram for the pair-wise Bonferroni–Dunn Test.

B.3. Experimental comparison of different margins

On the modified CIFAR-10 dataset, we try two different margins during our novel cDCM loss model training, which are 5 and 10. In Table 10, AUCROC values in these two margins are almost same. In addition, we can see in some normal classes, like automobile, deer, and etc., the F_2 measure on margin 5 is slightly better than the F_2 measure on margin 10, while in some other normal classes, like dog, frog, and etc., the F_2 measure on margin 10 is slightly better than the F_2 measure on margin 5. There is no clear evidence to show which margin is the best, but choosing different margins would not affect the final results too much. Based on our novel cDCM loss function, setting the margin too small results in a tiny hypersphere, while setting the margin too big results in a huge hypersphere. In the worst scenario, setting margin to 0 or infinity is meaningless. Hence, hyper-parameter "margin" should be set in a reasonable manner.

Appendix C. Tests of statistical significance

To verify whether our cDCM loss function has a significant difference from other loss functions on the modified CIFAR-10 dataset, we conduct a Friedman Test of statistical significance on the F_2 measure metric and the Bonferroni–Dunn Test for pair-wise statistical significance. Based on the Friedman test, we can reject the null hypothesis that there is no significance difference between the loss function at the 0.05 threshold (p-value is $9.34e^{-7}$).

The critical difference diagram for the pair-wise Bonferroni–Dunn Test in shown in Fig. 12. We can see the cDCM loss function is significantly better than BCE, WBCE and Focal loss functions, but the test shows no significant difference to the Deep SAD loss function at $\alpha = 0.05$.

Appendix D. Effect of decision threshold with the BCE loss applied to the modified CIFAR-10 dataset

Adjusting the decision threshold is a common technique for trading off precision and recall. Experimental results are shown in Tables 11 and 12. Although lowering the decision threshold from 0.5 with the BCE loss can gradually increase recall but the F_2 measure and accuracy does not improve to outperform cDCM on must classes. Even with the threshold set at an extremely small value of 0.01, the BCE loss still only outperforms cDCM on four classes: automobile, cat, dog, and ship. Furthermore, selecting such an extremely small decision threshold is problematic, and may not be optimal for the validation data. As illustrated in Fig. 6(a), there are many normal samples with prediction values greater than 0.01. Thus, simply moving the threshold of the BCE loss without prior knowledge of the entire testing data distribution cannot achieve better results than using cDCM.

Table 9
Comparison between different centers of our novel cDCM loss function on the modified CIFAR-10 dataset. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Center	Precision	Recall	F_2 measure	Accuracy	AUCROC
airplane	All 0.	0.9837 ± 0.0044	0.7264 ± 0.0456	0.7661 ± 0.0400	0.7427 ± 0.0377	0.8896 ± 0.0088
airplane	All 1.	0.9841 ± 0.0048	0.7239 ± 0.0594	0.7637 ± 0.0530	0.7407 ± 0.0496	0.8929 ± 0.0058
airplane	Random	0.9856 ± 0.0024	0.7147 ± 0.0357	0.7560 ± 0.0318	0.7338 ± 0.0304	0.8948 ± 0.0040
automobile	All 0.	0.9918 ± 0.0031	0.6888 ± 0.0318	0.7334 ± 0.0285	0.7147 ± 0.0265	0.8983 ± 0.0050
automobile	All 1.	0.9934 ± 0.0011	0.6735 ± 0.0172	0.7198 ± 0.0157	0.7021 ± 0.0151	0.8927 ± 0.0050
automobile	Random	0.9946 ± 0.0015	0.6568 ± 0.0397	0.7044 ± 0.0364	0.6879 ± 0.0347	0.8946 ± 0.0050
bird	All 0.	0.9530 ± 0.0074	0.6004 ± 0.0289	0.6482 ± 0.0267	0.6136 ± 0.0227	0.7225 ± 0.0247
bird	All 1.	0.9576 ± 0.0091	0.5641 ± 0.0552	0.6140 ± 0.0518	0.5849 ± 0.0436	0.7310 ± 0.0190
bird	Random	0.9548 ± 0.0037	0.5754 ± 0.0480	0.6246 ± 0.0449	0.5932 ± 0.0398	0.7261 ± 0.0138
cat	All 0.	0.9703 ± 0.0043	0.5412 ± 0.0495	0.5932 ± 0.0470	0.5720 ± 0.0414	0.7725 ± 0.0126
cat	All 1.	0.9662 ± 0.0038	0.6053 ± 0.0479	0.6538 ± 0.0448	0.6256 ± 0.0396	0.7741 ± 0.0056
cat	Random	0.9745 ± 0.0020	0.5273 ± 0.0376	0.5803 ± 0.0362	0.5621 ± 0.0323	0.7786 ± 0.0145
deer	All 0.	0.9711 ± 0.0068	0.6908 ± 0.0627	0.7324 ± 0.0558	0.7029 ± 0.0504	0.8378 ± 0.0062
deer	All 1.	0.9728 ± 0.0018	0.6546 ± 0.0418	0.7002 ± 0.0385	0.6727 ± 0.0358	0.8261 ± 0.0058
deer	Random	0.9750 ± 0.0033	0.6275 ± 0.0374	0.6754 ± 0.0345	0.6502 ± 0.0311	0.8310 ± 0.0074
dog	All 0.	0.9821 ± 0.0020	0.6717 ± 0.0213	0.7169 ± 0.0192	0.6934 ± 0.0177	0.8688 ± 0.0009
dog	All 1.	0.9852 ± 0.0033	0.6231 ± 0.0402	0.6722 ± 0.0372	0.6522 ± 0.0337	0.8624 ± 0.0043
dog	Random	0.9840 ± 0.0030	0.6415 ± 0.0490	0.6891 ± 0.0451	0.6678 ± 0.0417	0.8645 ± 0.0032
frog	All 0.	0.9880 ± 0.0067	0.7341 ± 0.0384	0.7736 ± 0.0335	0.7526 ± 0.0307	0.9124 ± 0.0186
frog	All 1.	0.9909 ± 0.0016	0.7090 ± 0.0142	0.7518 ± 0.0126	0.7323 ± 0.0117	0.9107 ± 0.0056
frog	Random	0.9887 ± 0.0048	0.7477 ± 0.0571	0.7855 ± 0.0502	0.7651 ± 0.0476	0.9156 ± 0.0036
horse	All 0.	0.9863 ± 0.0020	0.6030 ± 0.0104	0.6538 ± 0.0098	0.6352 ± 0.0097	0.8604 ± 0.0144
horse	All 1.	0.9847 ± 0.0058	0.6120 ± 0.0689	0.6613 ± 0.0647	0.6420 ± 0.0583	0.8510 ± 0.0172
horse	Random	0.9868 ± 0.0036	0.5856 ± 0.0567	0.6368 ± 0.0536	0.6198 ± 0.0485	0.8489 ± 0.0176
ship	All 0.	0.9836 ± 0.0040	0.8684 ± 0.0263	0.8891 ± 0.0214	0.8685 ± 0.0203	0.9395 ± 0.0061
ship	All 1.	0.9880 ± 0.0028	0.8281 ± 0.0331	0.8557 ± 0.0280	0.8362 ± 0.0279	0.9412 ± 0.0062
ship	Random	0.9859 ± 0.0030	0.8618 ± 0.0303	0.8839 ± 0.0250	0.8645 ± 0.0245	0.9439 ± 0.0042
truck	All 0.	0.9865 ± 0.0015	0.7231 ± 0.0214	0.7638 ± 0.0191	0.7419 ± 0.0193	0.9019 ± 0.0085
truck	All 1.	0.9828 ± 0.0020	0.7610 ± 0.0298	0.7968 ± 0.0260	0.7729 ± 0.0250	0.8974 ± 0.0076
truck	Random	0.9830 ± 0.0032	0.7783 ± 0.0325	0.8119 ± 0.0281	0.7882 ± 0.0266	0.9063 ± 0.0063

Table 10
Comparison between different margins of our novel cDCM loss function on the modified CIFAR-10 dataset. We use the center of 'All 1.' here. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Margin	Precision	Recall	F_2 measure	Accuracy	AUCROC
airplane	5	0.9841 ± 0.0048	0.7239 ± 0.0594	0.7637 ± 0.0530	0.7407 ± 0.0496	0.8929 ± 0.0058
airplane	10	0.9823 ± 0.0030	0.7328 ± 0.0438	0.7717 ± 0.0391	0.7476 ± 0.0374	0.8891 ± 0.0093
automobile	5	0.9934 ± 0.0011	0.6735 ± 0.0172	0.7198 ± 0.0157	0.7021 ± 0.0151	0.8927 ± 0.0050
automobile	10	0.9943 ± 0.0017	0.6500 ± 0.0197	0.6983 ± 0.0180	0.6816 ± 0.0167	0.8951 ± 0.0078
bird	5	0.9576 ± 0.0091	0.5641 ± 0.0552	0.6140 ± 0.0518	0.5849 ± 0.0436	0.7310 ± 0.0190
bird	10	0.9549 ± 0.0049	0.5776 ± 0.0710	0.6263 ± 0.0653	0.5951 ± 0.0582	0.7323 ± 0.0132
cat	5	0.9662 ± 0.0038	0.6053 ± 0.0479	0.6538 ± 0.0448	0.6256 ± 0.0396	0.7741 ± 0.0056
cat	10	0.9623 ± 0.0089	0.6230 ± 0.0595	0.6695 ± 0.0540	0.6382 ± 0.0460	0.7730 ± 0.0114
deer	5	0.9728 ± 0.0018	0.6546 ± 0.0418	0.7002 ± 0.0385	0.6727 ± 0.0358	0.8261 ± 0.0058
deer	10	0.9754 ± 0.0032	0.6280 ± 0.0452	0.6758 ± 0.0411	0.6508 ± 0.0378	0.8308 ± 0.0107
dog	5	0.9852 ± 0.0033	0.6231 ± 0.0402	0.6722 ± 0.0372	0.6522 ± 0.0337	0.8624 ± 0.0043
dog	10	0.9828 ± 0.0004	0.6580 ± 0.0118	0.7046 ± 0.0108	0.6818 ± 0.0105	0.8652 ± 0.0051
frog	5	0.9909 ± 0.0016	0.7090 ± 0.0142	0.7518 ± 0.0126	0.7323 ± 0.0117	0.9107 ± 0.0056
frog	10	0.9867 ± 0.0017	0.7602 ± 0.0222	0.7967 ± 0.0195	0.7750 ± 0.1920	0.9123 ± 0.0073
horse	5	0.9847 ± 0.0058	0.6120 ± 0.0689	0.6613 ± 0.0647	0.6420 ± 0.0583	0.8510 ± 0.0172
horse	10	0.9884 ± 0.0058	0.5724 ± 0.0549	0.6245 ± 0.0516	0.6089 ± 0.0459	0.8606 ± 0.0109
ship	5	0.9880 ± 0.0028	0.8281 ± 0.0331	0.8557 ± 0.0280	0.8362 ± 0.0279	0.9412 ± 0.0062
ship	10	0.9870 ± 0.0055	0.8448 ± 0.0460	0.8695 ± 0.0378	0.8501 ± 0.0364	0.9465 ± 0.0026
truck	5	0.9828 ± 0.0020	0.7610 ± 0.0298	0.7968 ± 0.0260	0.7729 ± 0.0250	0.8974 ± 0.0076
truck	10	0.9785 ± 0.0041	0.8023 ± 0.0297	0.8321 ± 0.0251	0.8061 ± 0.0241	0.9030 ± 0.0110

Table 11
Comparison between the cDCM loss function and the binary cross-entropy(BCE) loss function with various thresholds on the modified CIFAR-10 dataset. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Loss	Threshold	Precision	Recall	F_2 measure	Accuracy	AUCROC
airplane	cDCM		0.9841 ± 0.0048	0.7239 ± 0.0594	0.7637 ± 0.0530	0.7407 ± 0.0496	0.8929 ± 0.0058
airplane	BCE	0.01	0.9833 ± 0.0038	0.7227 ± 0.0554	0.7626 ± 0.0486	0.7392 ± 0.0463	0.8812 ± 0.0041
airplane	BCE	0.05	0.9887 ± 0.0035	0.6438 ± 0.0547	0.6915 ± 0.0497	0.6726 ± 0.0464	0.8812 ± 0.0041
airplane	BCE	0.1	0.9907 ± 0.0026	0.6040 ± 0.0539	0.6546 ± 0.0498	0.6383 ± 0.0464	0.8812 ± 0.0041
airplane	BCE	0.2	0.9925 ± 0.0014	0.5593 ± 0.0516	0.6123 ± 0.0489	0.5995 ± 0.0455	0.8812 ± 0.0041
airplane	BCE	0.3	0.9933 ± 0.0007	0.5293 ± 0.0508	0.5834 ± 0.0488	0.5731 ± 0.0451	0.8812 ± 0.0041
airplane	BCE	0.5	0.9942 ± 0.0008	0.4796 ± 0.0509	0.5346 ± 0.0500	0.5292 ± 0.0455	0.8812 ± 0.0041
airplane	BCE	0.7	0.9946 ± 0.0003	0.4275 ± 0.0502	0.4821 ± 0.0505	0.4827 ± 0.0450	0.8812 ± 0.0041
automobile	cDCM		0.9934 ± 0.0011	0.6735 ± 0.0172	0.7198 ± 0.0157	0.7021 ± 0.0150	0.8927 ± 0.0050
automobile	BCE	0.01	0.9907 ± 0.0029	0.7271 ± 0.0354	**0.7678** ± 0.0311	0.7482 ± 0.0299	0.9075 ± 0.0056
automobile	BCE	0.05	0.9944 ± 0.0013	0.6594 ± 0.0365	0.7068 ± 0.0334	0.6901 ± 0.0320	0.9075 ± 0.0056
automobile	BCE	0.1	0.9954 ± 0.0009	0.6260 ± 0.0378	0.6759 ± 0.0351	0.6608 ± 0.0334	0.9075 ± 0.0056
automobile	BCE	0.2	0.9964 ± 0.0007	0.5912 ± 0.0397	0.6433 ± 0.0376	0.6301 ± 0.0353	0.9075 ± 0.0056
automobile	BCE	0.3	0.9968 ± 0.0006	0.5668 ± 0.0428	0.6200 ± 0.0410	0.6085 ± 0.0383	0.9075 ± 0.0056
automobile	BCE	0.5	0.9974 ± 0.0003	0.5278 ± 0.0446	0.5824 ± 0.0435	0.5738 ± 0.0400	0.9075 ± 0.0056
automobile	BCE	0.7	0.9979 ± 0.0005	0.4881 ± 0.0475	0.5433 ± 0.0472	0.5384 ± 0.0427	0.9075 ± 0.0056
bird	cDCM		0.9576 ± 0.0091	0.5641 ± 0.0552	0.6140 ± 0.0518	0.5849 ± 0.0436	0.7310 ± 0.0190
bird	BCE	0.01	0.9701 ± 0.0034	0.3962 ± 0.0332	0.4491 ± 0.0340	0.4455 ± 0.0279	0.7356 ± 0.0057
bird	BCE	0.05	0.9737 ± 0.0024	0.3125 ± 0.0232	0.3616 ± 0.0248	0.3736 ± 0.0198	0.7356 ± 0.0057
bird	BCE	0.1	0.9757 ± 0.0007	0.2767 ± 0.0212	0.3229 ± 0.0230	0.3428 ± 0.0186	0.7356 ± 0.0057
bird	BCE	0.2	0.9788 ± 0.0009	0.2412 ± 0.0193	0.2839 ± 0.0213	0.3123 ± 0.0168	0.7356 ± 0.0057
bird	BCE	0.3	0.9798 ± 0.0015	0.2198 ± 0.0175	0.2601 ± 0.0195	0.2937 ± 0.0154	0.7356 ± 0.0057
bird	BCE	0.5	0.9832 ± 0.0016	0.1861 ± 0.0173	0.2221 ± 0.0195	0.2646 ± 0.0151	0.7356 ± 0.0057
bird	BCE	0.7	0.9835 ± 0.0023	0.1548 ± 0.0156	0.1862 ± 0.0180	0.2369 ± 0.0137	0.7356 ± 0.0057
cat	cDCM		0.9662 ± 0.0038	0.6053 ± 0.0479	0.6538 ± 0.0448	0.6256 ± 0.0396	0.7741 ± 0.0056
cat	BCE	0.01	0.9529 ± 0.0104	0.6610 ± 0.1550	**0.6998** ± 0.1454	0.6642 ± 0.1274	0.7503 ± 0.0111
cat	BCE	0.05	0.9638 ± 0.0063	0.5295 ± 0.1380	0.5783 ± 0.1367	0.5580 ± 0.1170	0.7503 ± 0.0111
cat	BCE	0.1	0.9685 ± 0.0059	0.4657 ± 0.1259	0.5166 ± 0.1279	0.5050 ± 0.1077	0.7503 ± 0.0111
cat	BCE	0.2	0.9729 ± 0.0054	0.3994 ± 0.1113	0.4503 ± 0.1158	0.4490 ± 0.0957	0.7503 ± 0.0111
cat	BCE	0.3	0.9756 ± 0.0047	0.3546 ± 0.1017	0.4042 ± 0.1075	0.4108 ± 0.0877	0.7503 ± 0.0111
cat	BCE	0.5	0.9791 ± 0.0035	0.2883 ± 0.0884	0.3340 ± 0.0956	0.3537 ± 0.0770	0.7503 ± 0.0111
cat	BCE	0.7	0.9843 ± 0.0026	0.2301 ± 0.0749	0.2706 ± 0.0827	0.3036 ± 0.0657	0.7503 ± 0.0111
deer	cDCM		0.9728 ± 0.0018	0.6546 ± 0.0418	0.7002 ± 0.0385	0.6727 ± 0.0358	0.8261 ± 0.0058
deer	BCE	0.01	0.9731 ± 0.0075	0.6317 ± 0.0757	0.6784 ± 0.0682	0.6524 ± 0.0613	0.8208 ± 0.0033
deer	BCE	0.05	0.9811 ± 0.0059	0.5096 ± 0.0998	0.5620 ± 0.0958	0.5494 ± 0.0850	0.8208 ± 0.0033
deer	BCE	0.1	0.9837 ± 0.0056	0.4581 ± 0.1110	0.5106 ± 0.1095	0.5050 ± 0.0955	0.8208 ± 0.0033
deer	BCE	0.2	0.9868 ± 0.0067	0.4032 ± 0.1161	0.4546 ± 0.1180	0.4574 ± 0.1002	0.8208 ± 0.0033
deer	BCE	0.3	0.9881 ± 0.0063	0.3679 ± 0.1194	0.4179 ± 0.1236	0.4266 ± 0.1037	0.8208 ± 0.0033
deer	BCE	0.5	0.9908 ± 0.0060	0.3201 ± 0.1235	0.3671 ± 0.1312	0.3848 ± 0.1082	0.8208 ± 0.0033
deer	BCE	0.7	0.9921 ± 0.0052	0.2761 ± 0.1249	0.3194 ± 0.1355	0.3460 ± 0.1101	0.8208 ± 0.0033
dog	cDCM		0.9852 ± 0.0033	0.6231 ± 0.0402	0.6722 ± 0.0372	0.6522 ± 0.0337	0.8624 ± 0.0043
dog	BCE	0.01	0.9749 ± 0.0087	0.6695 ± 0.0777	**0.7131** ± 0.0697	0.6865 ± 0.0626	0.8348 ± 0.0049
dog	BCE	0.05	0.9821 ± 0.0064	0.5857 ± 0.0709	0.6362 ± 0.0666	0.6172 ± 0.0593	0.8348 ± 0.0049
dog	BCE	0.1	0.9845 ± 0.0046	0.5473 ± 0.0663	0.5999 ± 0.0637	0.5846 ± 0.0565	0.8348 ± 0.0049
dog	BCE	0.2	0.9875 ± 0.0046	0.5058 ± 0.0613	0.5598 ± 0.0601	0.5493 ± 0.0527	0.8348 ± 0.0049
dog	BCE	0.3	0.9895 ± 0.0043	0.4786 ± 0.0581	0.5330 ± 0.0577	0.5259 ± 0.0502	0.8348 ± 0.0049
dog	BCE	0.5	0.9917 ± 0.0033	0.4332 ± 0.0545	0.4877 ± 0.0552	0.4865 ± 0.0475	0.8348 ± 0.0049
dog	BCE	0.7	0.9936 ± 0.0028	0.3921 ± 0.0526	0.4456 ± 0.0542	0.4505 ± 0.0460	0.8348 ± 0.0049

Table 12
Comparison between the cDCM loss function and the binary cross-entropy(BCE) loss function with various thresholds on the modified CIFAR-10 dataset. Each result value in the table is the average of 5 experiments with different weight initialization.

Normal class	Loss	Threshold	Precision	Recall	F_2 measure	Accuracy	AUCROC
frog	cDCM		0.9909 ± 0.0016	0.7090 ± 0.0142	0.7518 ± 0.0126	0.7323 ± 0.0117	0.9107 ± 0.0056
frog	BCE	0.01	0.9900 ± 0.0033	0.6591 ± 0.0608	0.7057 ± 0.0557	0.6871 ± 0.0523	0.8822 ± 0.0097
frog	BCE	0.05	0.9937 ± 0.0023	0.5850 ± 0.0648	0.6367 ± 0.0612	0.6230 ± 0.0568	0.8822 ± 0.0097
frog	BCE	0.1	0.9946 ± 0.0020	0.5532 ± 0.0669	0.6062 ± 0.0642	0.5951 ± 0.0591	0.8822 ± 0.0097
frog	BCE	0.2	0.9957 ± 0.0014	0.5184 ± 0.0691	0.5725 ± 0.0673	0.5645 ± 0.0614	0.8822 ± 0.0097
frog	BCE	0.3	0.9963 ± 0.0012	0.4961 ± 0.0701	0.5507 ± 0.0690	0.5448 ± 0.0625	0.8822 ± 0.0097
frog	BCE	0.5	0.9972 ± 0.0012	0.4610 ± 0.0710	0.5156 ± 0.0709	0.5137 ± 0.0633	0.8822 ± 0.0097
frog	BCE	0.7	0.9978 ± 0.0010	0.4270 ± 0.0717	0.4812 ± 0.0728	0.4834 ± 0.0641	0.8822 ± 0.0097
horse	cDCM		0.9847 ± 0.0058	0.6120 ± 0.0689	0.6613 ± 0.0647	0.6420 ± 0.0583	0.8510 ± 0.0172
horse	BCE	0.01	0.9890 ± 0.0014	0.5868 ± 0.0255	0.6386 ± 0.0241	0.6222 ± 0.0221	0.8685 ± 0.0054
horse	BCE	0.05	0.9930 ± 0.0013	0.4895 ± 0.0143	0.5447 ± 0.0142	0.5374 ± 0.0126	0.8685 ± 0.0054
horse	BCE	0.1	0.9946 ± 0.0016	0.4500 ± 0.0138	0.5053 ± 0.0139	0.5028 ± 0.0120	0.8685 ± 0.0054
horse	BCE	0.2	0.9953 ± 0.0015	0.4093 ± 0.0144	0.4638 ± 0.0149	0.4666 ± 0.0125	0.8685 ± 0.0054
horse	BCE	0.3	0.9958 ± 0.0013	0.3855 ± 0.0152	0.4393 ± 0.0158	0.4454 ± 0.0134	0.8685 ± 0.0054
horse	BCE	0.5	0.9971 ± 0.0006	0.3479 ± 0.0155	0.4000 ± 0.0164	0.4122 ± 0.0138	0.8685 ± 0.0054
horse	BCE	0.7	0.9982 ± 0.0007	0.3129 ± 0.0169	0.3627 ± 0.0181	0.3811 ± 0.0151	0.8685 ± 0.0054
ship	cDCM		0.9880 ± 0.0028	0.8281 ± 0.0331	0.8557 ± 0.0280	0.8362 ± 0.0279	0.9412 ± 0.0062
ship	BCE	0.01	0.9731 ± 0.0091	0.8963 ± 0.0655	**0.9099** ± 0.0538	0.8839 ± 0.0508	0.9309 ± 0.0051
ship	BCE	0.05	0.9852 ± 0.0041	0.8187 ± 0.0644	0.8467 ± 0.0551	0.8256 ± 0.0543	0.9309 ± 0.0051
ship	BCE	0.1	0.9888 ± 0.0030	0.7729 ± 0.0666	0.8075 ± 0.0577	0.7876 ± 0.0571	0.9309 ± 0.0051
ship	BCE	0.2	0.9912 ± 0.0027	0.7160 ± 0.0774	0.7571 ± 0.0685	0.7385 ± 0.0672	0.9309 ± 0.0051
ship	BCE	0.3	0.9925 ± 0.0027	0.6744 ± 0.0872	0.7194 ± 0.0787	0.7022 ± 0.0761	0.9309 ± 0.0051
ship	BCE	0.5	0.9947 ± 0.0023	0.6066 ± 0.1057	0.6561 ± 0.0988	0.6429 ± 0.0933	0.9309 ± 0.0051
ship	BCE	0.7	0.9957 ± 0.0022	0.5374 ± 0.1232	0.5892 ± 0.1193	0.5814 ± 0.1093	0.9309 ± 0.0051
truck	cDCM		0.9828 ± 0.0020	0.7610 ± 0.0298	0.7968 ± 0.0260	0.7729 ± 0.0250	0.8974 ± 0.0076
truck	BCE	0.01	0.9897 ± 0.0030	0.6564 ± 0.0792	0.7027 ± 0.0732	0.6844 ± 0.0688	0.8924 ± 0.0078
truck	BCE	0.05	0.9931 ± 0.0022	0.5710 ± 0.0707	0.6232 ± 0.0683	0.6102 ± 0.0622	0.8924 ± 0.0078
truck	BCE	0.1	0.9943 ± 0.0015	0.5317 ± 0.0684	0.5854 ± 0.0673	0.5757 ± 0.0605	0.8924 ± 0.0078
truck	BCE	0.2	0.9956 ± 0.0013	0.4891 ± 0.0656	0.5438 ± 0.0659	0.5382 ± 0.0584	0.8924 ± 0.0078
truck	BCE	0.3	0.9964 ± 0.0011	0.4605 ± 0.0626	0.5153 ± 0.0639	0.5129 ± 0.0558	0.8924 ± 0.0078
truck	BCE	0.5	0.9971 ± 0.0008	0.4183 ± 0.0595	0.4726 ± 0.0620	0.4753 ± 0.0532	0.8924 ± 0.0078
truck	BCE	0.7	0.9979 ± 0.0006	0.3748 ± 0.0571	0.4276 ± 0.0606	0.4365 ± 0.0512	0.8924 ± 0.0078

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